

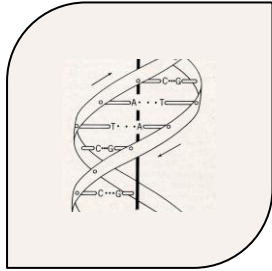
NATURAL SELECTION AND ADAPTATION

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EVOLUTIONARY PROCESSES



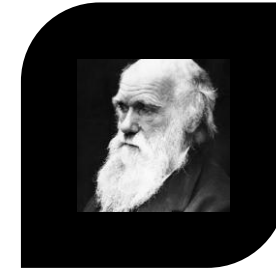
MUTATION



GENE FLOW

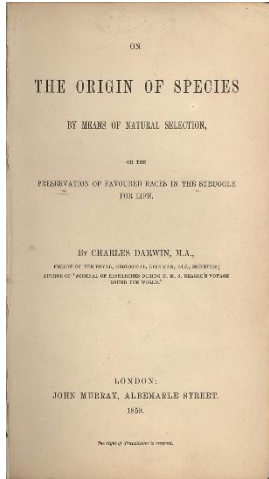


DRIFT

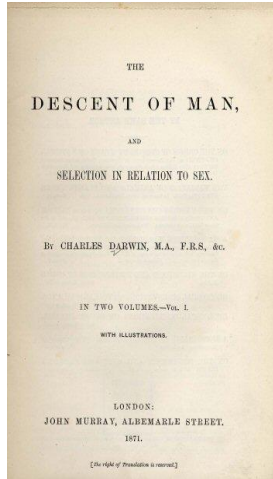


NATURAL
SELECTION

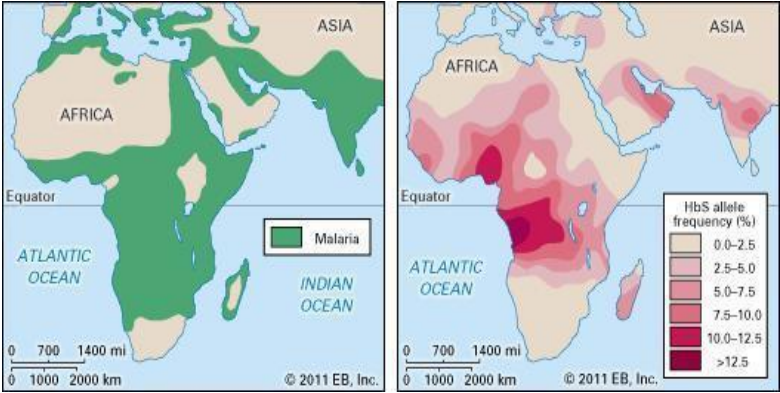
NATURAL SELECTION



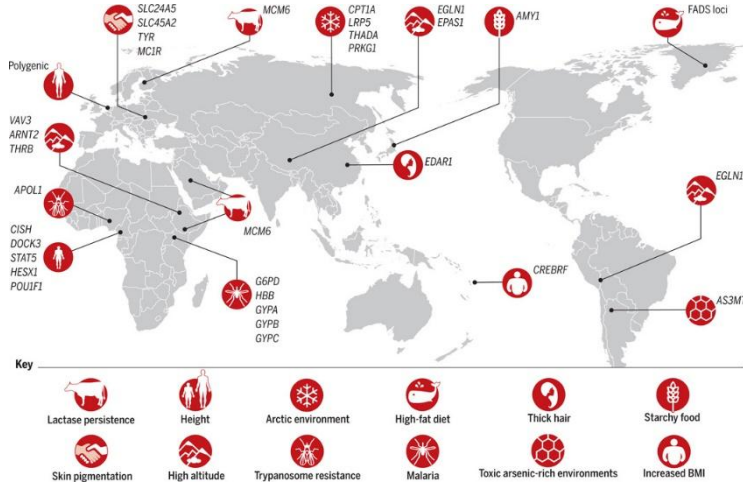
1859



1871



1949

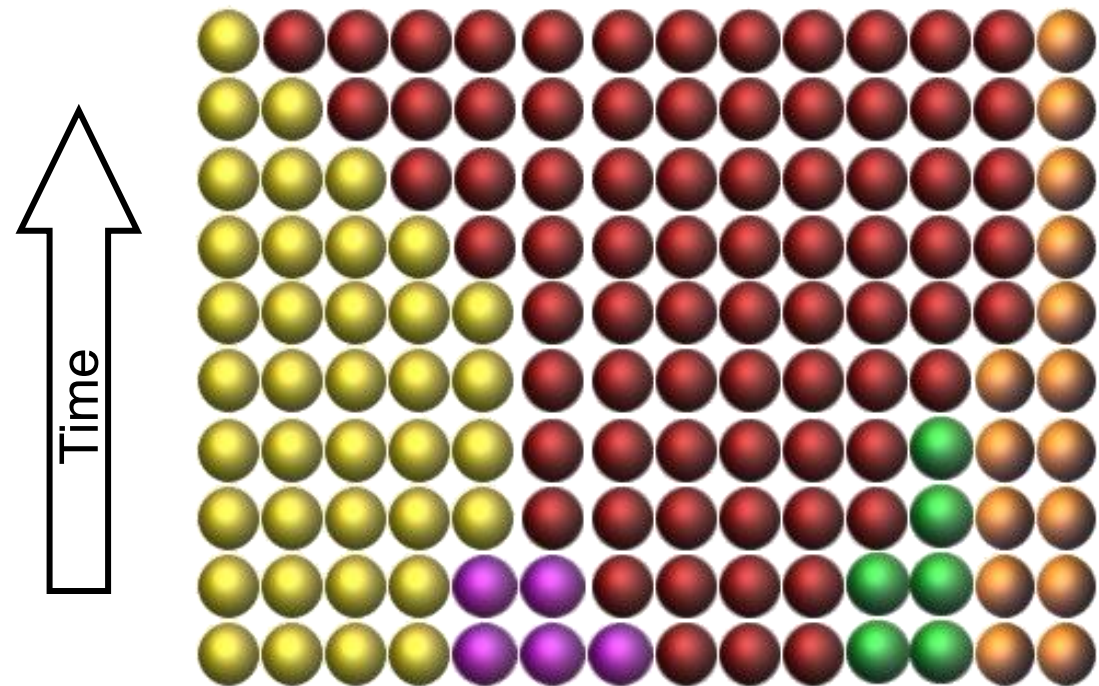


2000

CHARLES DARWIN

- i) Populations have a phenotypic variation.
- ii) The environment presents challenges.
- iii) Those better able to cope tend to leave more offspring.
- iv) Individuals tend to produce more offspring than the environment can support.

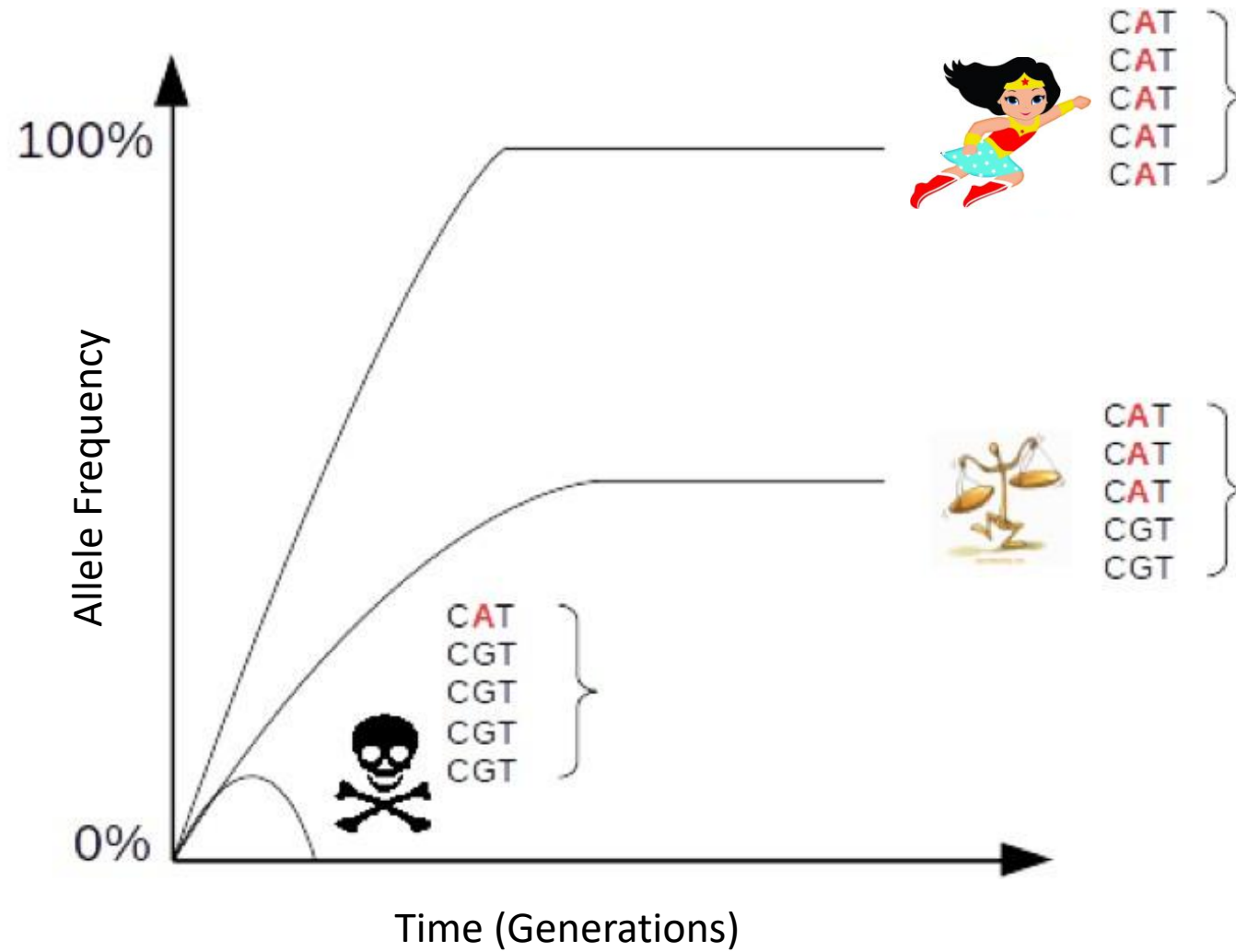
- i) Variation.
- ii) Variation must contribute to survival and differential reproduction.
- iii) Variation must be inherited.



Natural Selection

- Heritable traits that increase fitness become more common.
- Sites targeted by natural selection are likely to harbor functionality.
- Mutations arise (almost) randomly and evolve according to their effect on the carrier's fitness.

Natural Selection



Allele Frequency Trajectories

Effect of Selection on Alleles:

- Neutral/Weak: removed, polymorphic or fixed;
- Strong Negative: Removed or polymorphic;
- Strong Positive: removed, polymorphic or fixed.
- Balancing: removed, polymorphic or fixed.

Allele Frequency Trajectories

Effect of Selection on Alleles:

- Neutral/Weak: removed, polymorphic or fixed;
- Strong Negative: Removed or polymorphic;
- Strong Positive: removed, polymorphic or fixed.
- Balancing: removed, polymorphic or fixed.

What is strong? It depends on the effective population size.

Allele Frequency is (frequently) not enough to determine selection.

Allele Frequency Trajectories

If the simple observation of allele frequencies is not enough, what else can we do to detect signals of natural selection?

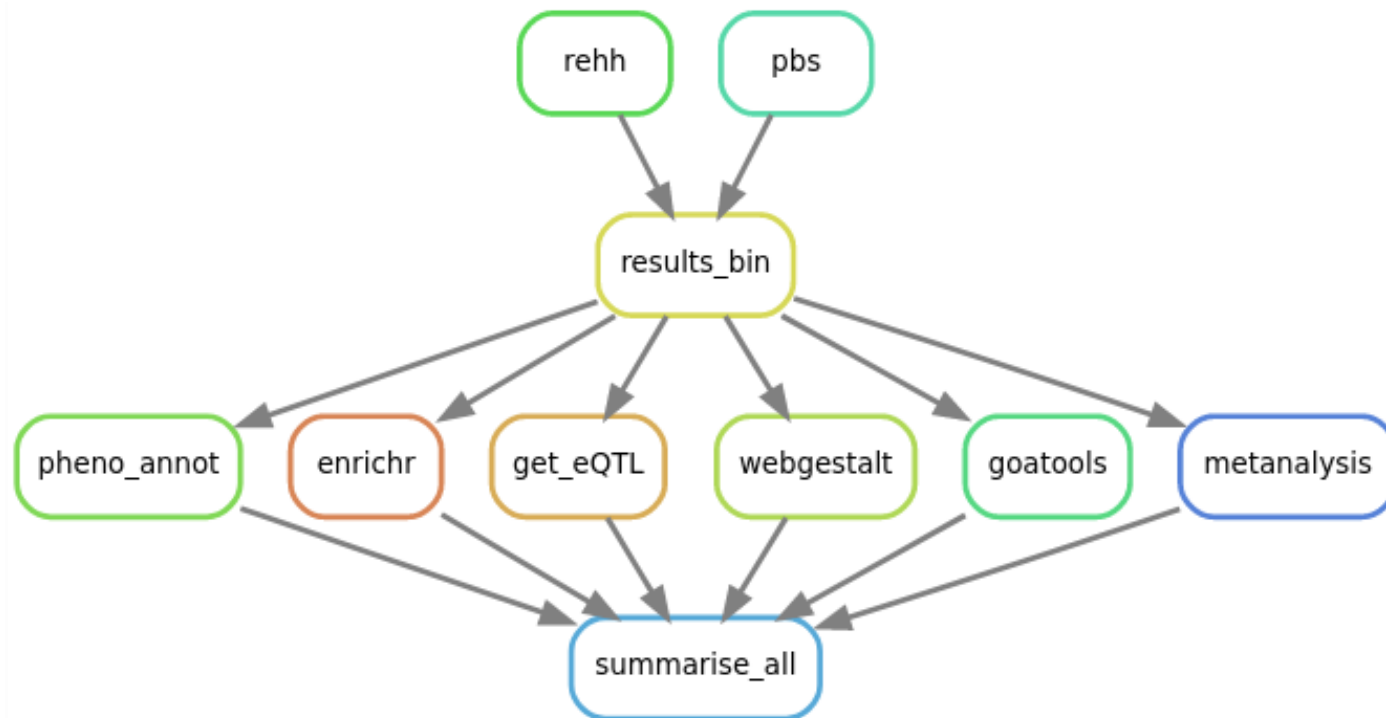
Allele Frequency Trajectories

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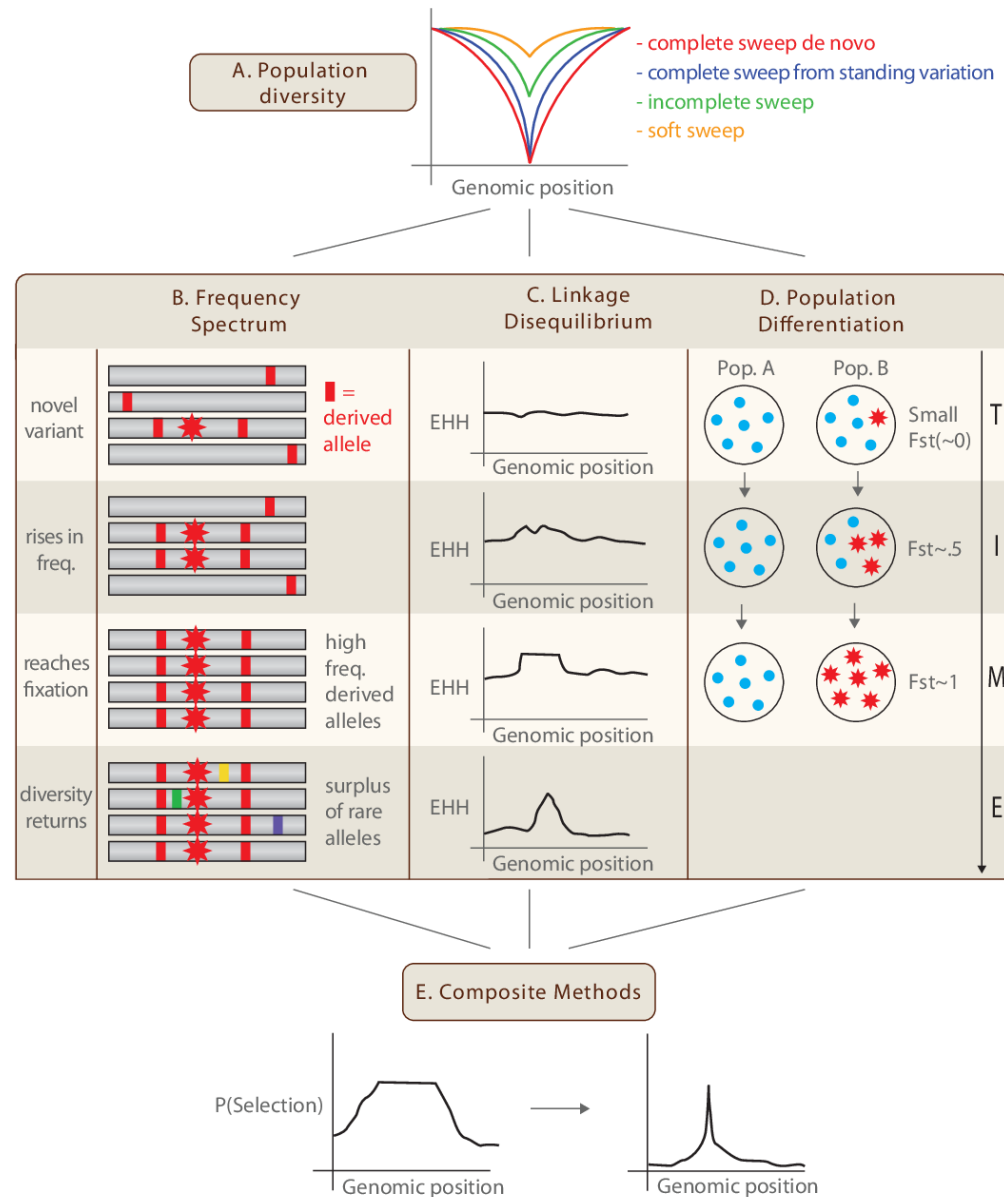
- perform selection experiments;
- use external information: candidate genes/biological knowledge, functional categories, association to phenotypes;
- use information from the surrounding genomic region;
- use information from multiple species/populations;

Allele Frequency Trajectories

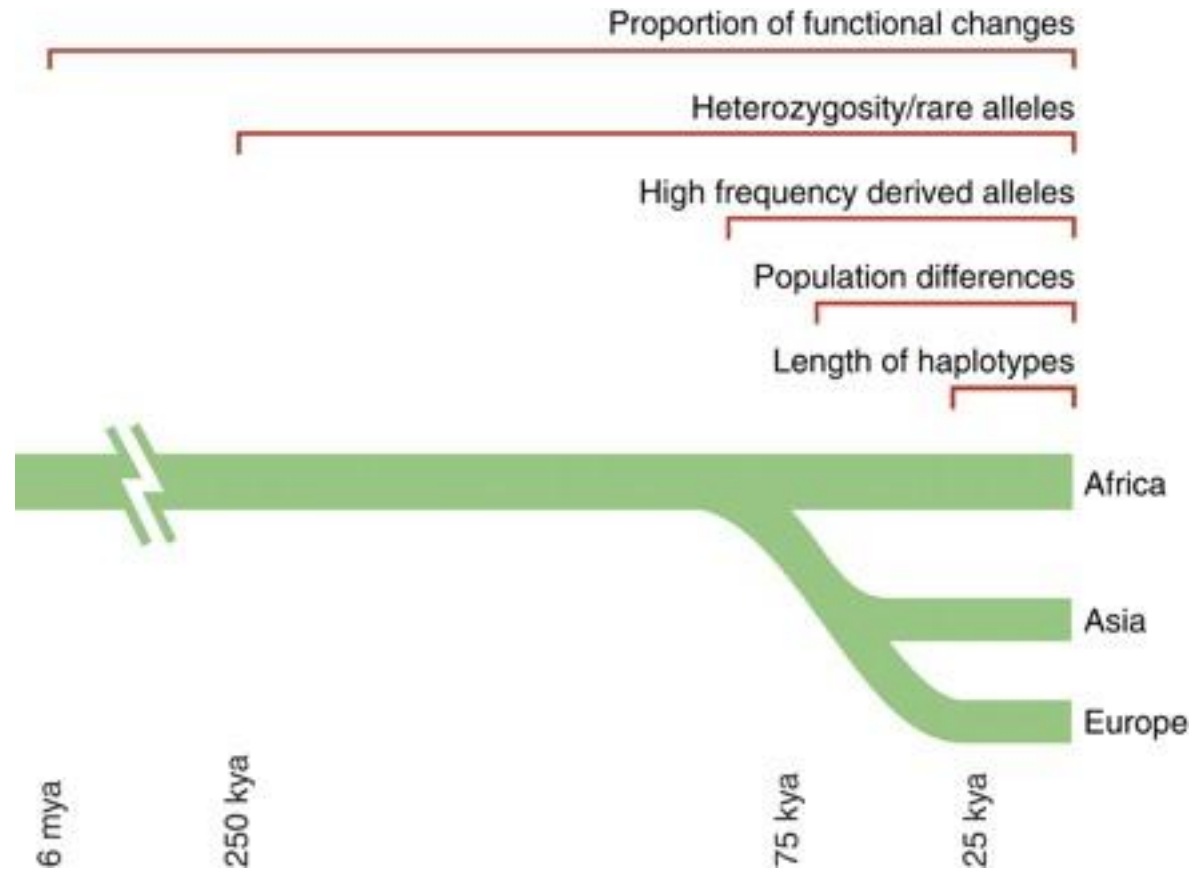
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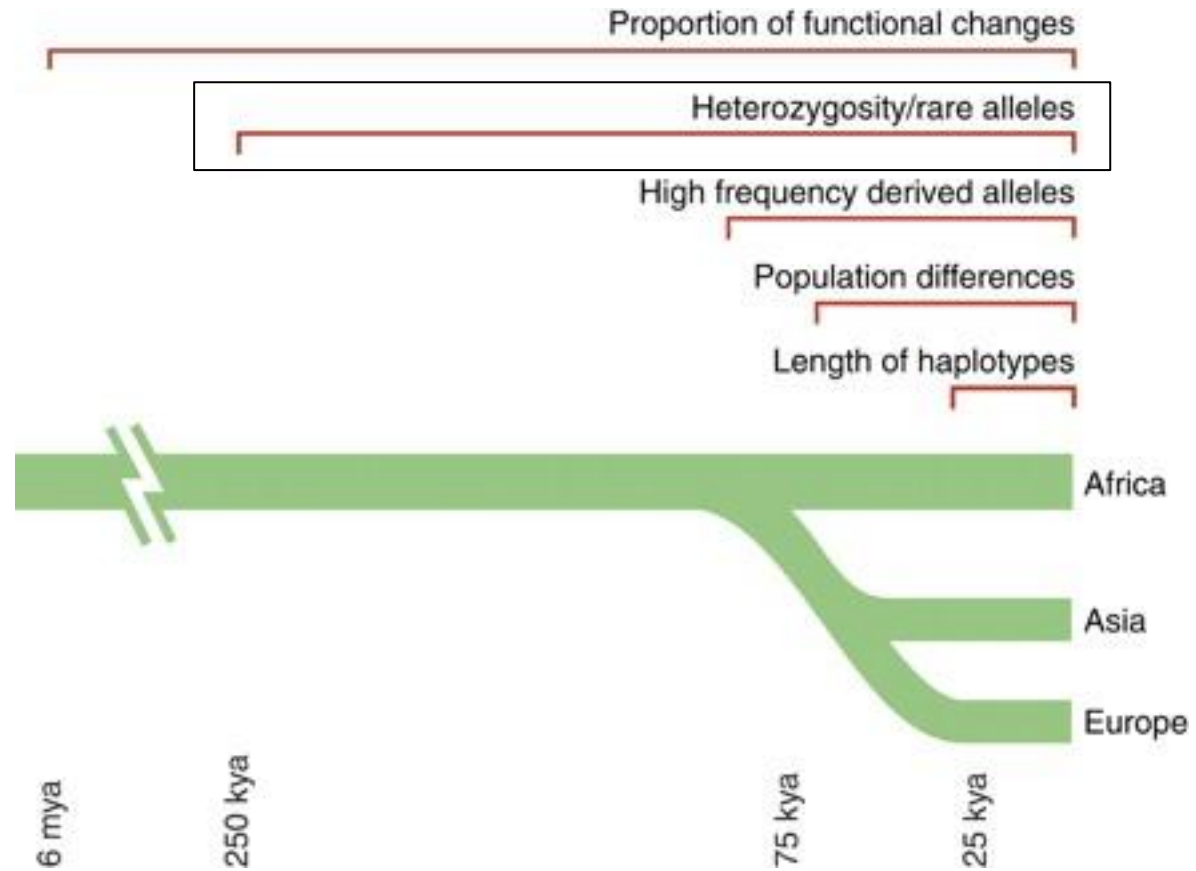
Common Methods to Detect Selection



Common Methods to Detect Selection



Common Methods to Detect Selection



Tajima's D

ACTAGAGGAT
ACTTGAGGAT
ACTTGACGAT
ACTTGACGAT
ACTTGAGGTT

$$\hat{\theta}_T = \frac{\sum_{i < j} d_{ij}}{n(n-1)/2}$$

$$\hat{\theta}_W = \frac{S}{n-1} \frac{1}{\sum_{i=1}^{n-1} 1/i}$$

$$D = \frac{\hat{\theta}_T - \hat{\theta}_W}{\sqrt{\hat{V}(\hat{\theta}_T - \hat{\theta}_W)}}$$

Tajima's D measures how many standard deviations the difference between π and θ_W departs from the expectation under neutrality.

Z-score

Tajima's D

ACTAGAGGAT
ACTTGAGGAT
ACTTGACGAT
ACTTGACGAT
ACTTGAGGTT

$$\hat{\theta}_T = \frac{\sum_{i < j} d_{ij}}{n(n-1)/2}$$

$$\hat{\theta}_W = \frac{S}{n-1} \frac{1}{\sum_{i=1} 1/i}$$

$$D = \frac{\hat{\theta}_T - \hat{\theta}_W}{\sqrt{\hat{V}(\hat{\theta}_T - \hat{\theta}_W)}}$$

$\Theta_T = \pi$ = average number of differences between pairs of sequences

$$\Theta_T = 1+2+2+2+1+1+1+0+2/10 = 1.2$$

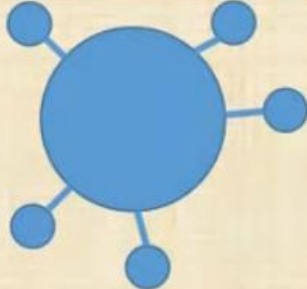
$\Theta_W = \pi$ expected number under neutrality

$$\Theta_W = 3/(1/1+1/2+1/3+1/4)=1.44$$

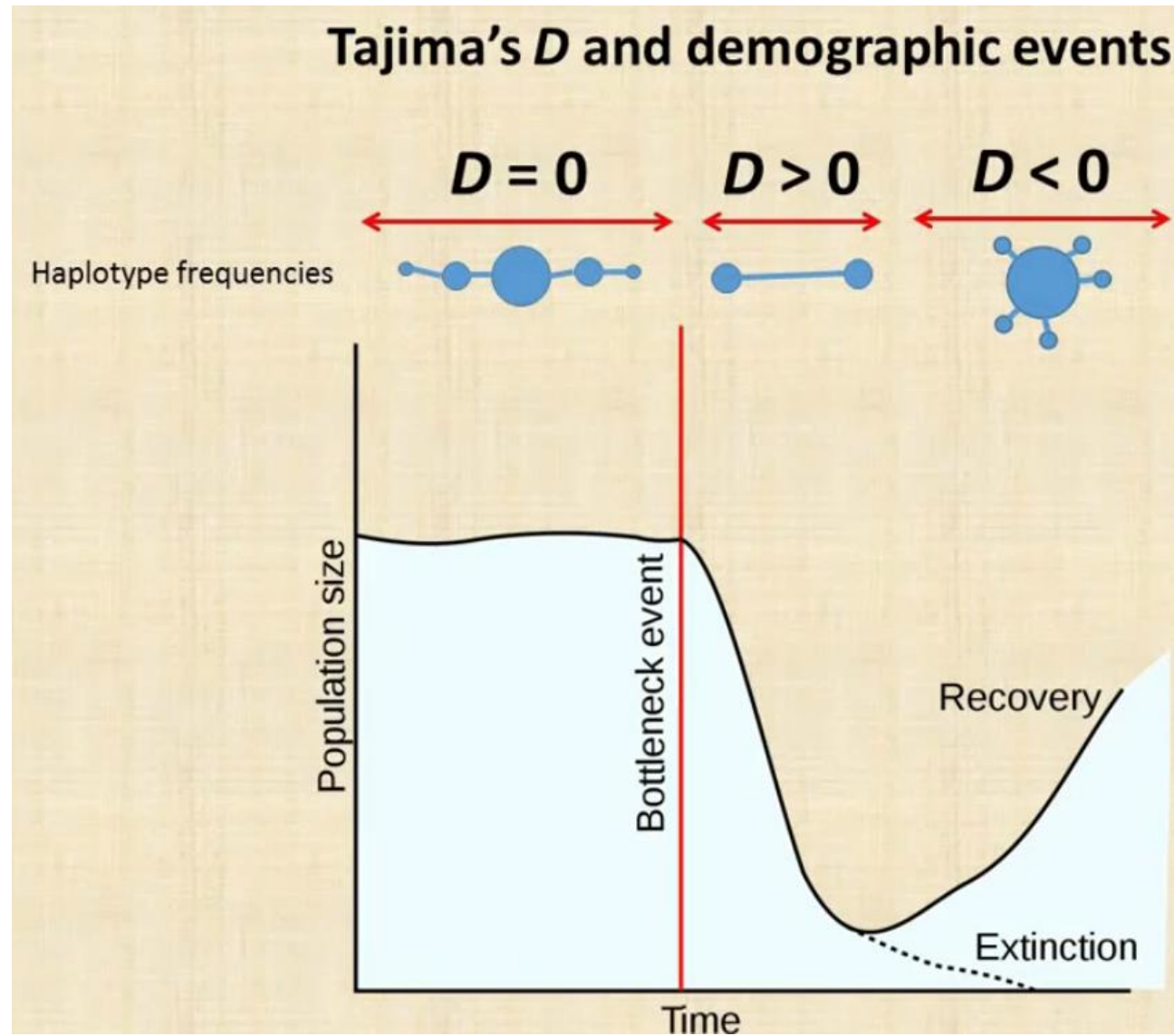
Tajima's D

Tajima's D

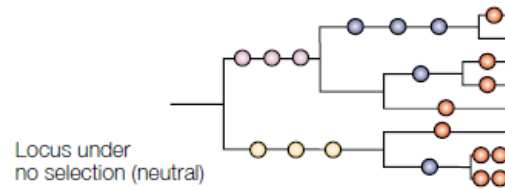
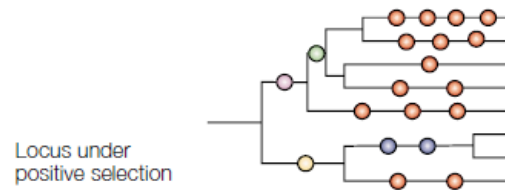
$D \approx \text{observed genetic variation} - \text{expected genetic variation for a given number of individuals}$

Tajima's D = 0 Observed and expected genetic variation given the pop size are the same Population at equilibrium A main haplotype and several derived ones (normal distribution)	 A haplotype network diagram showing a central large blue circle connected to four smaller blue circles, representing a main haplotype and several derived ones in a normal distribution.
Tajima's D < 0 Lower genetic variation than expected given the population size One haplotype dominates with rare nearly identical ones (rare alleles overrepresented)	 A haplotype network diagram showing a central large blue circle connected to six smaller blue circles, representing one dominant haplotype and several rare, nearly identical ones (rare alleles overrepresented).
Tajima's D > 0 Higher genetic variation than expected given the population size No main haplotypes, rather several unrelated haplotypes with similar frequency coexists (no rare alleles)	 A haplotype network diagram showing two large blue circles connected by a line, representing two unrelated haplotypes with similar frequency coexisting (no rare alleles).

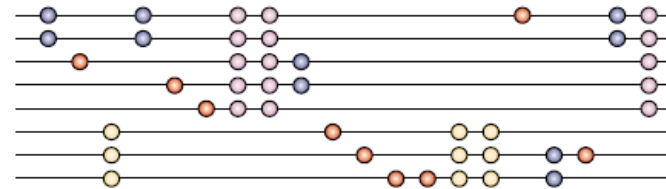
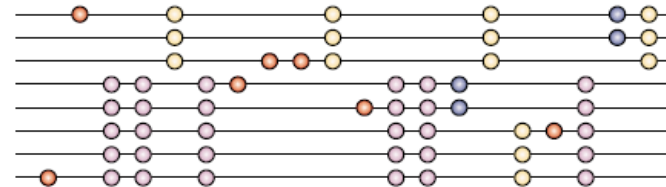
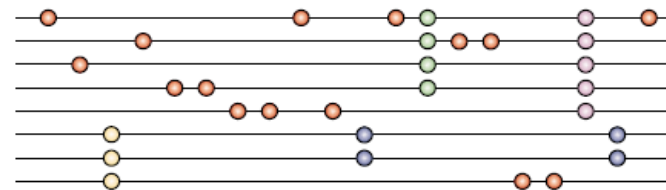
Tajima's D



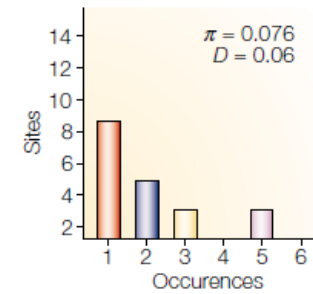
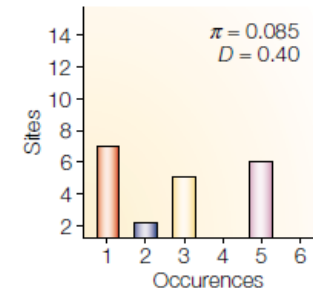
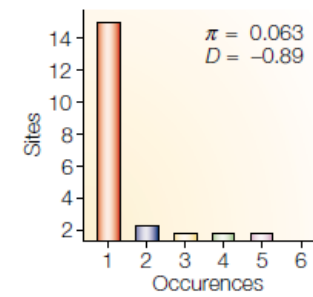
a Genealogies



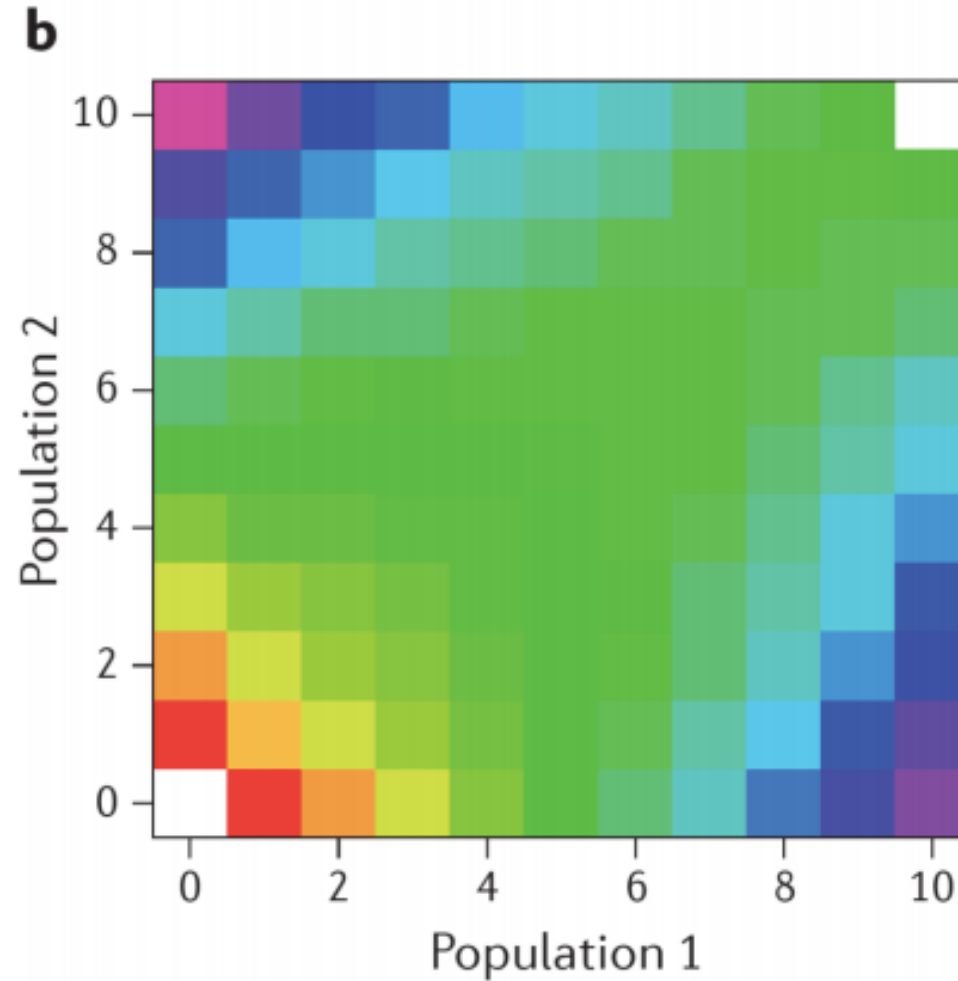
b Haplotypes



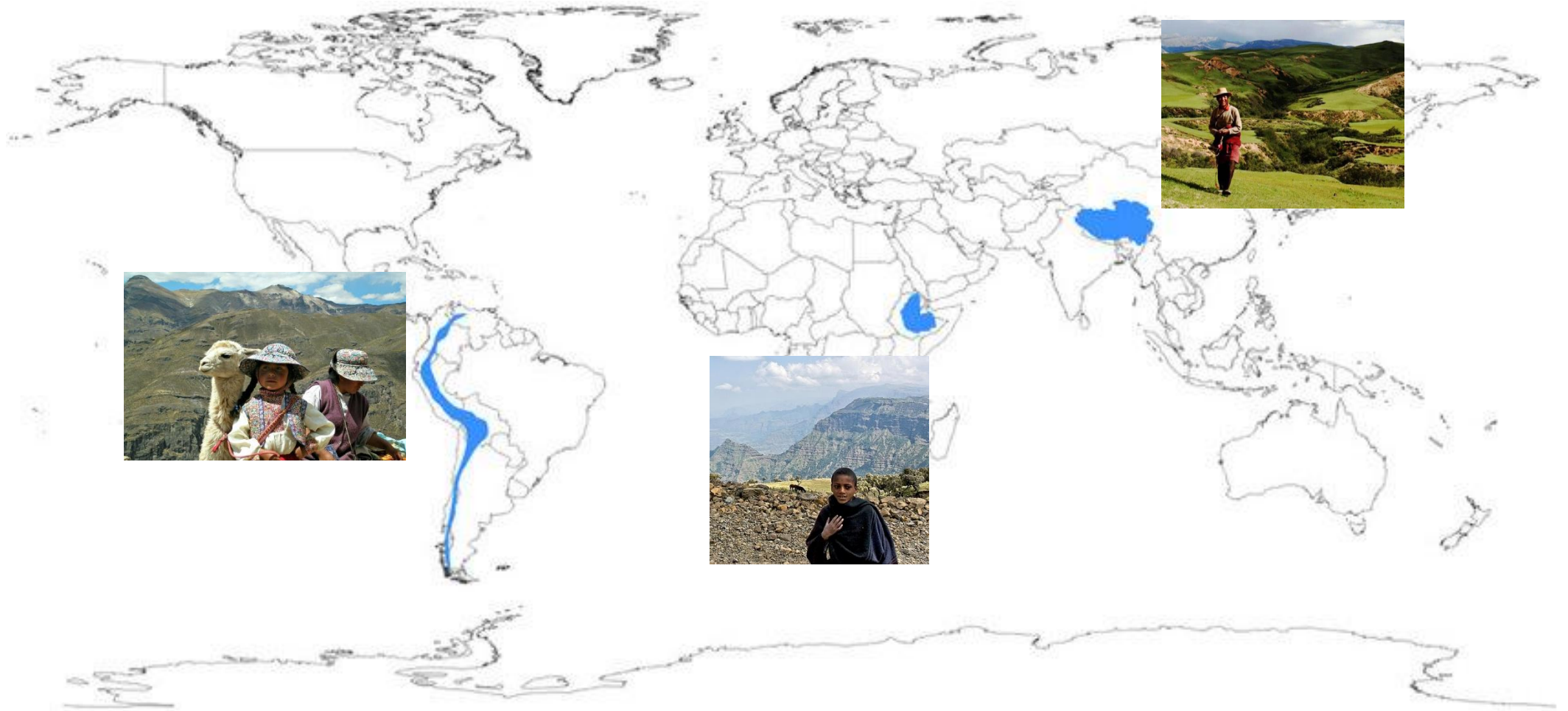
c Site frequency spectra



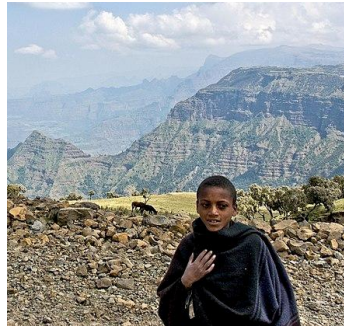
- For a pair of populations – 2D SFS
 - Count the SNPs have a frequency of the derived allele of i in population 1, and of j in population 2
- We can extend this to 3D SFS, 4D SFS, etc.



SFS/Natural Selection



SFS/Natural Selection

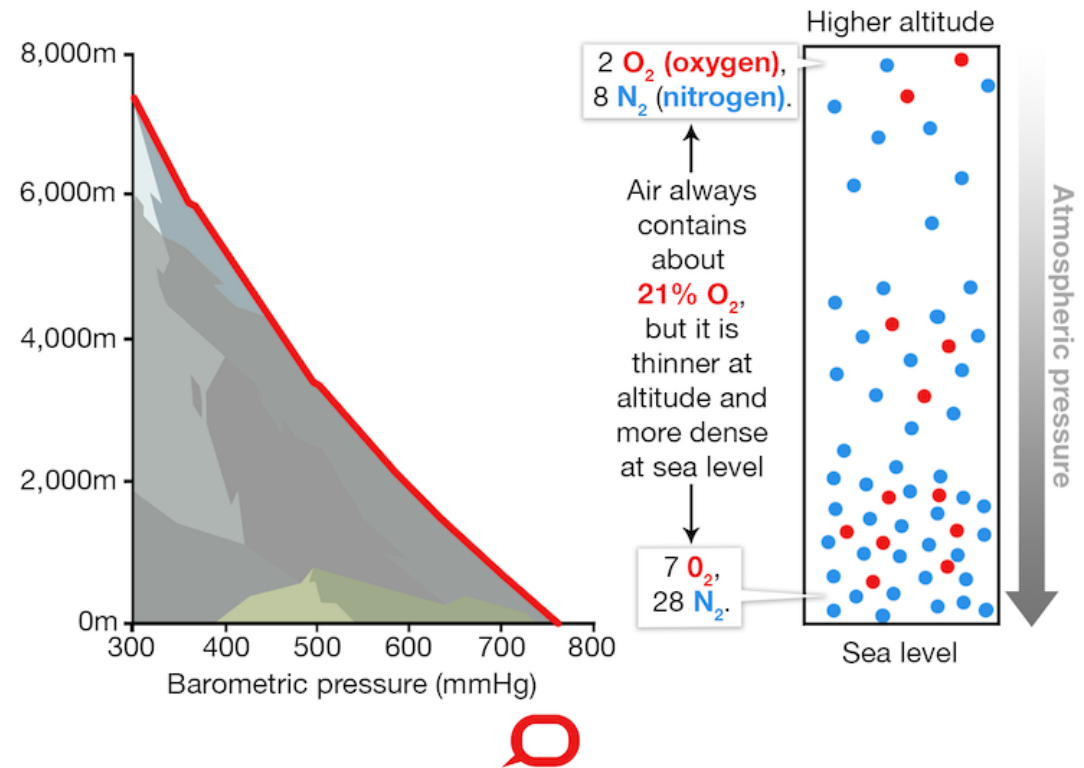


Low oxygen has large effect on fitness.

Birth defects.

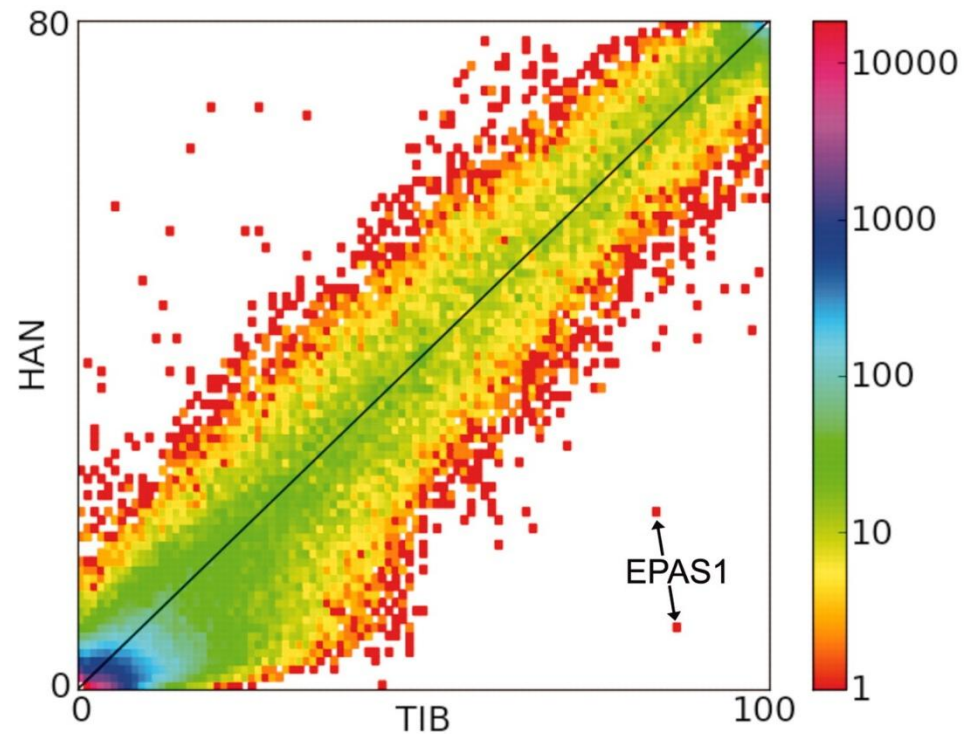
Problems during pregnancy.

The impact of altitude on oxygen levels

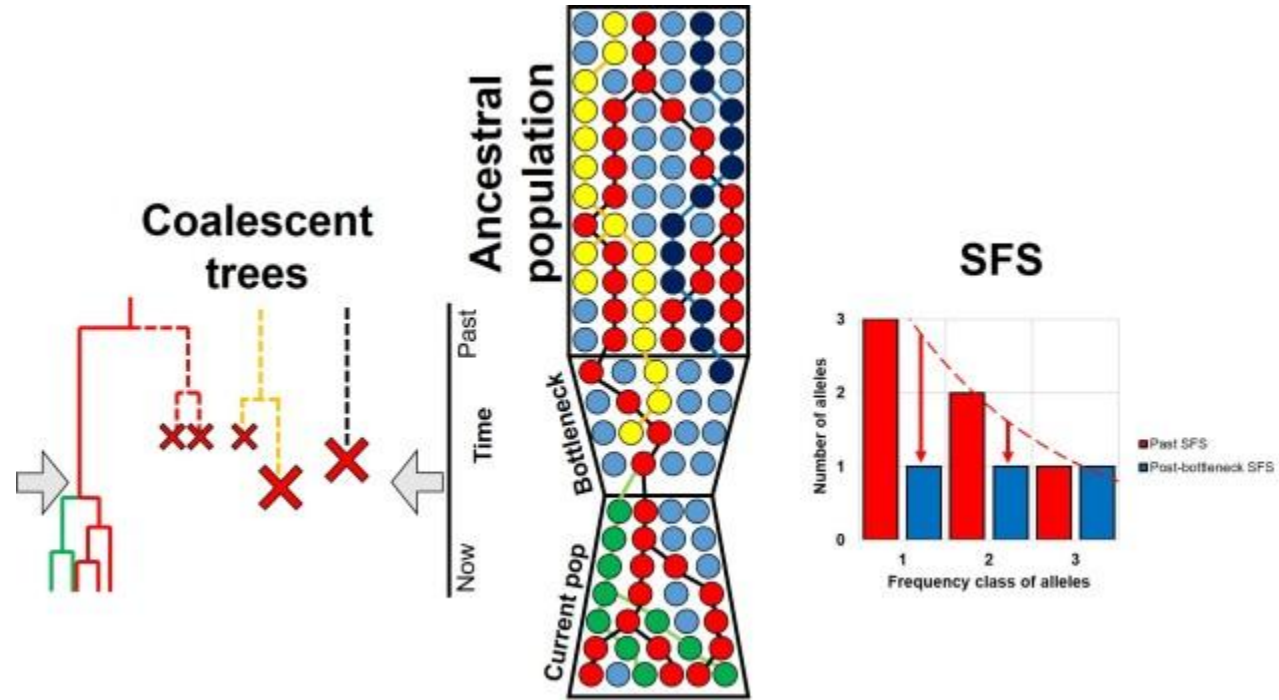


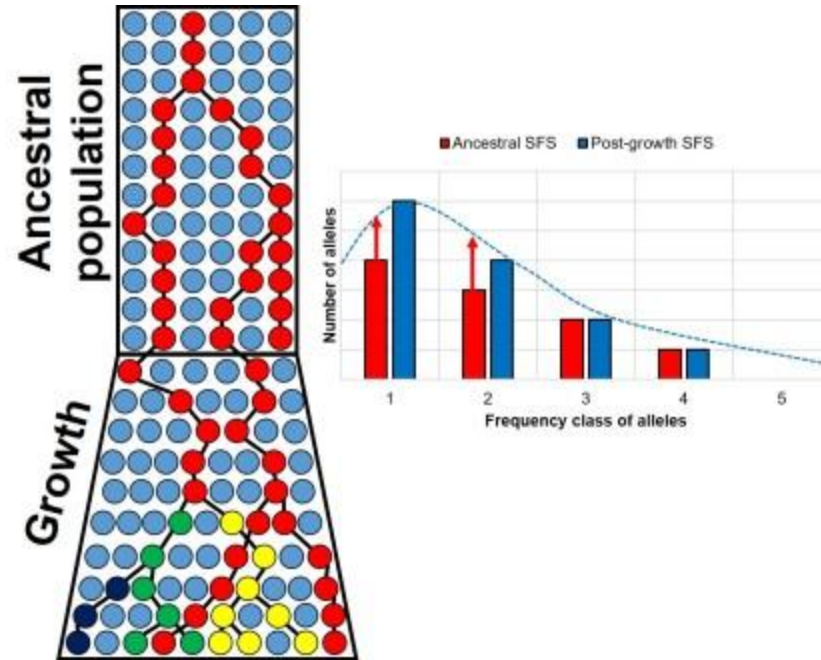
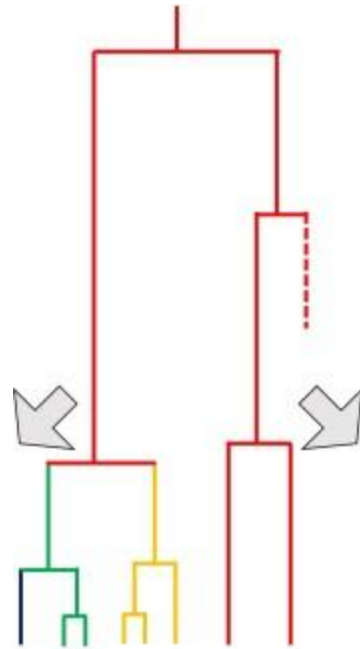
SFS/Natural Selection

- The full exomes of 50 Tibetan individuals at an average coverage of 18X.
- Compared to 40 Han Chinese individuals sequenced at an average of 6X (1000G).
- Estimated joint allele frequencies for each SNP using Bayesian approach.

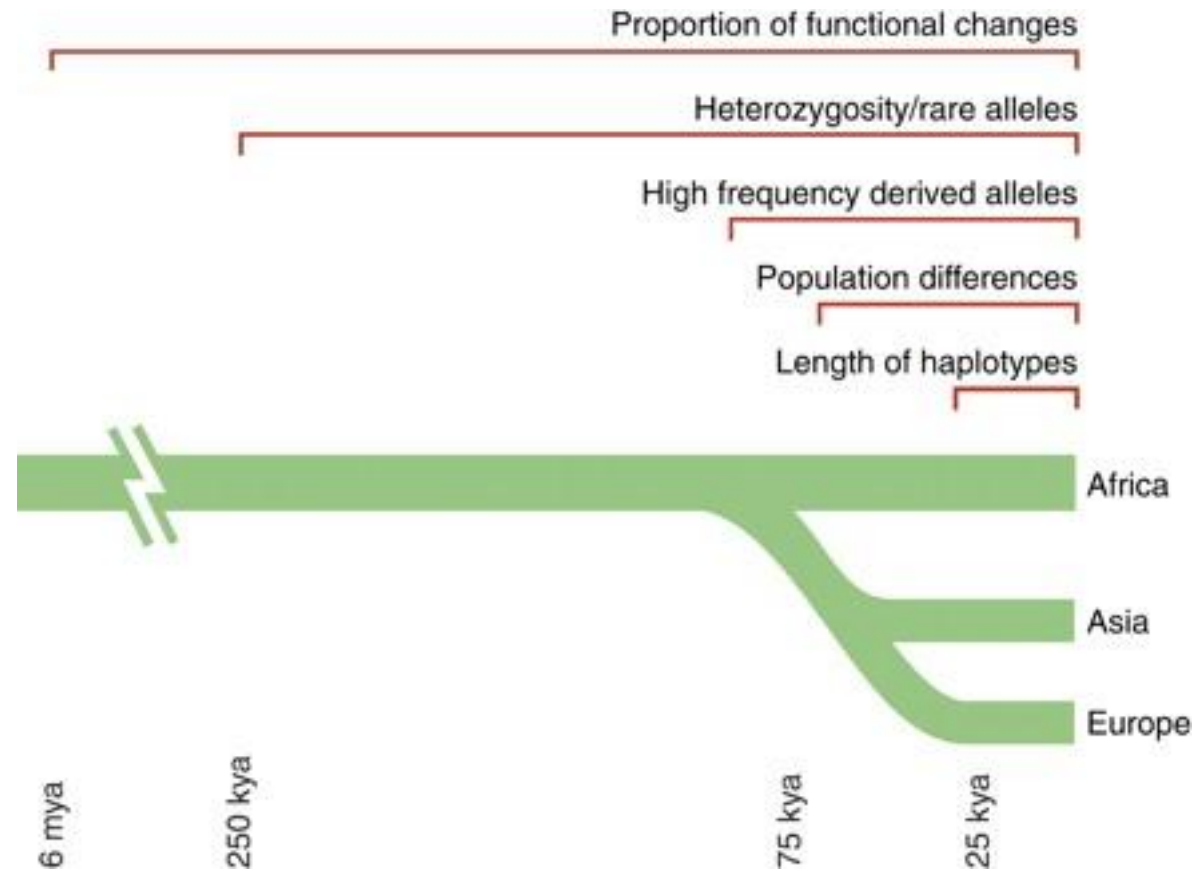


- i) **SNPs Bias:** Preselected SNP panels tend to include common variants, underestimating rare ones, distorting the frequency spectrum and potentially masking signals of selection.
- ii) **Admixture:** Gene flow between populations alters allele frequencies, creating excesses in certain frequency classes that can mimic or obscure selection signatures.
- iii) **Population Structure:** Subdivision within populations causes differences in allele frequencies between subgroups, producing spectrum deviations that may be mistaken for selection.
- iv) **Population Size:** Changes in population size (expansion or bottleneck) affect the frequency spectrum, potentially generating patterns similar to those expected under selection.



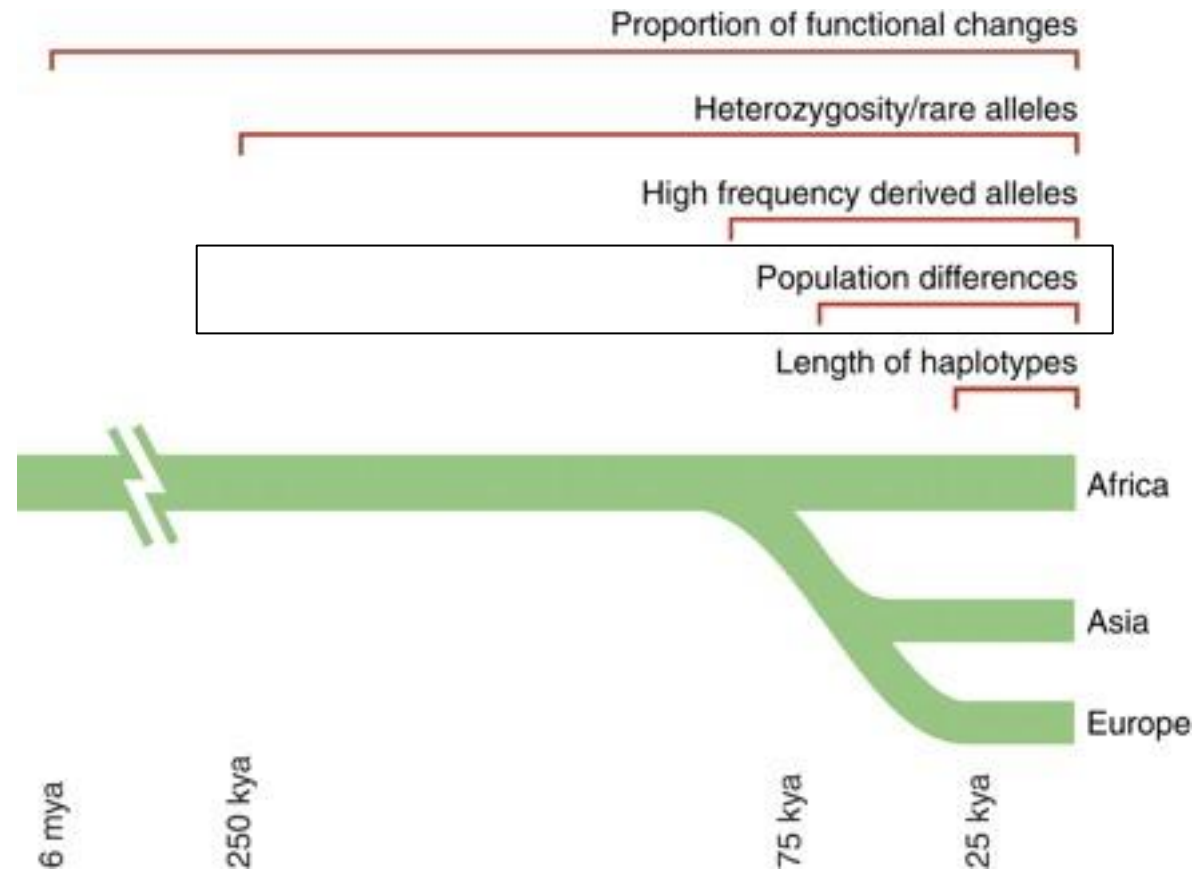


Common Methods to Detect Selection



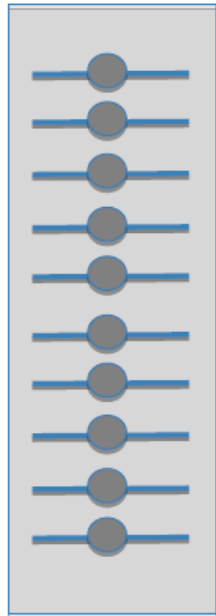
Sabeti PC, et al. Positive natural selection in the human lineage. *Science*. 2006 Jun 16;312(5780):1614-20. Review.

Common Methods to Detect Selection



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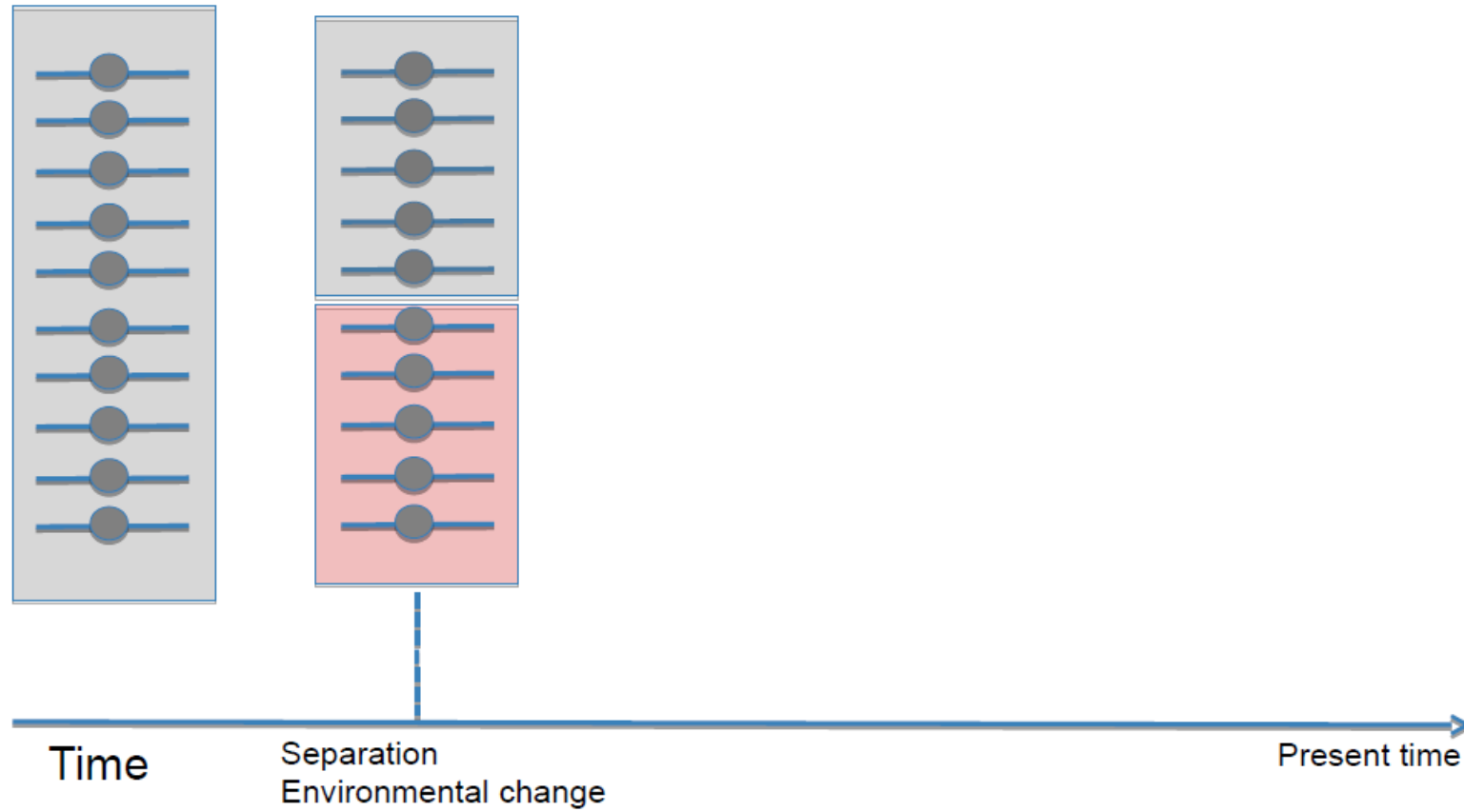
Allele Frequency Differentiation



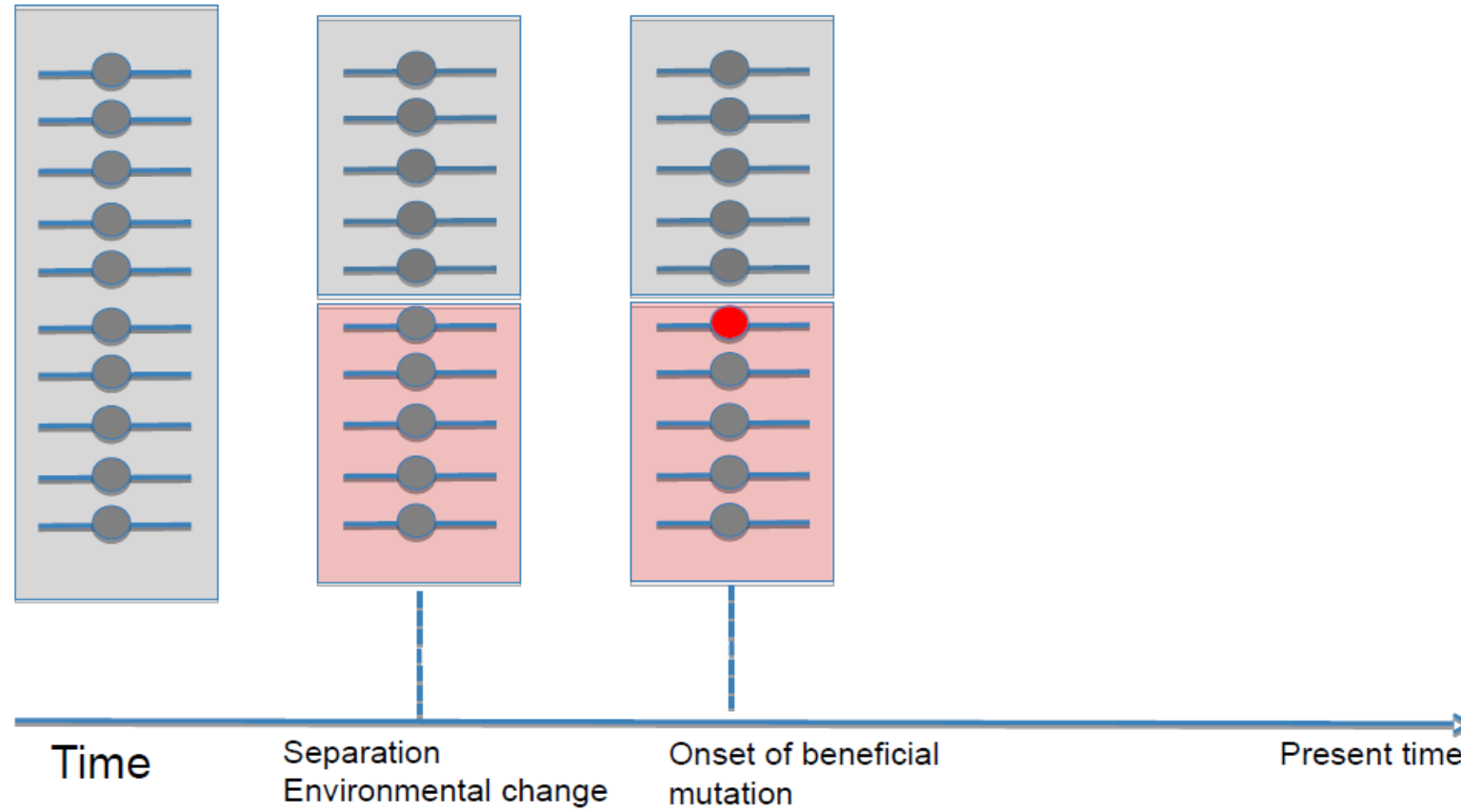
Time

Present time

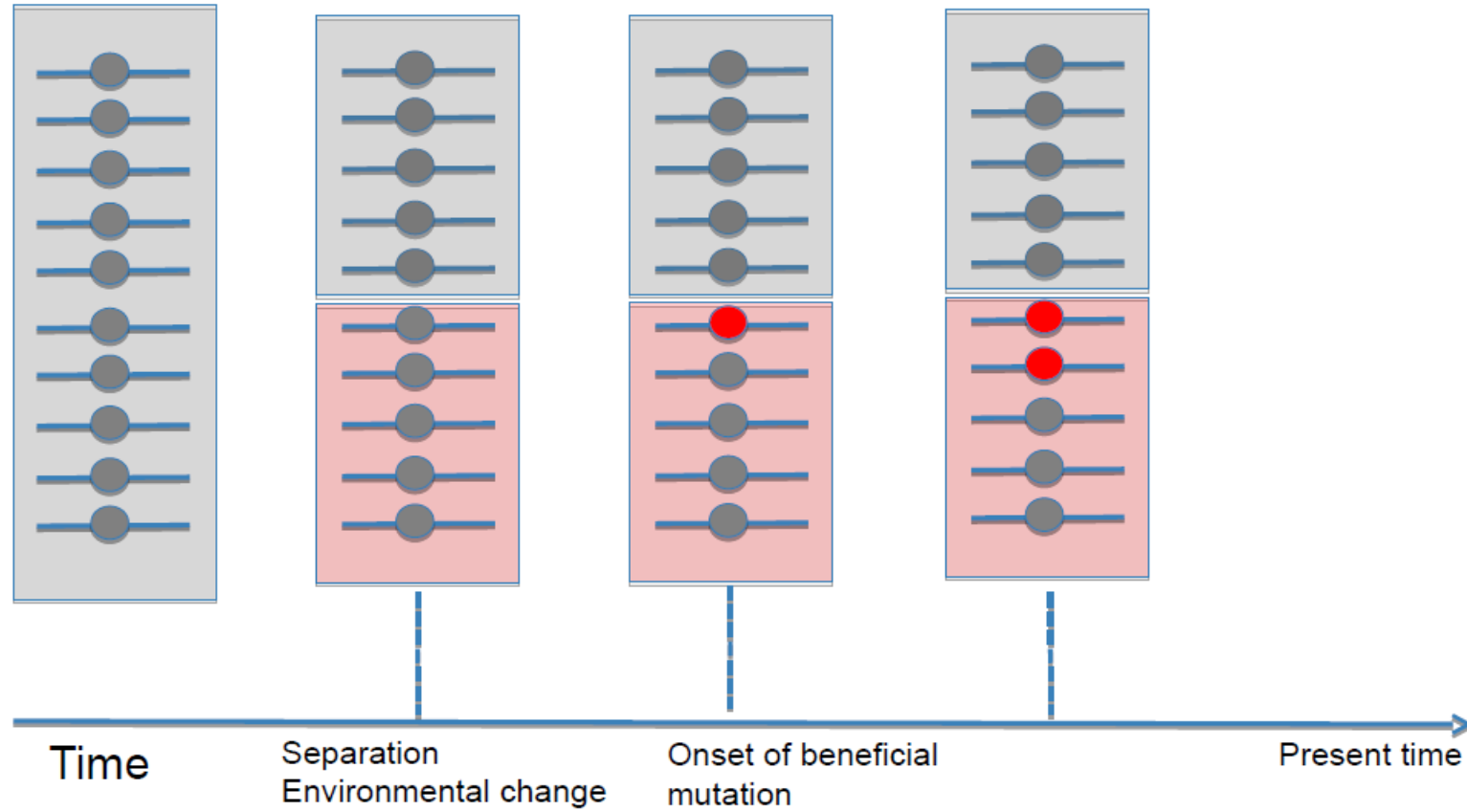
Allele Frequency Differentiation



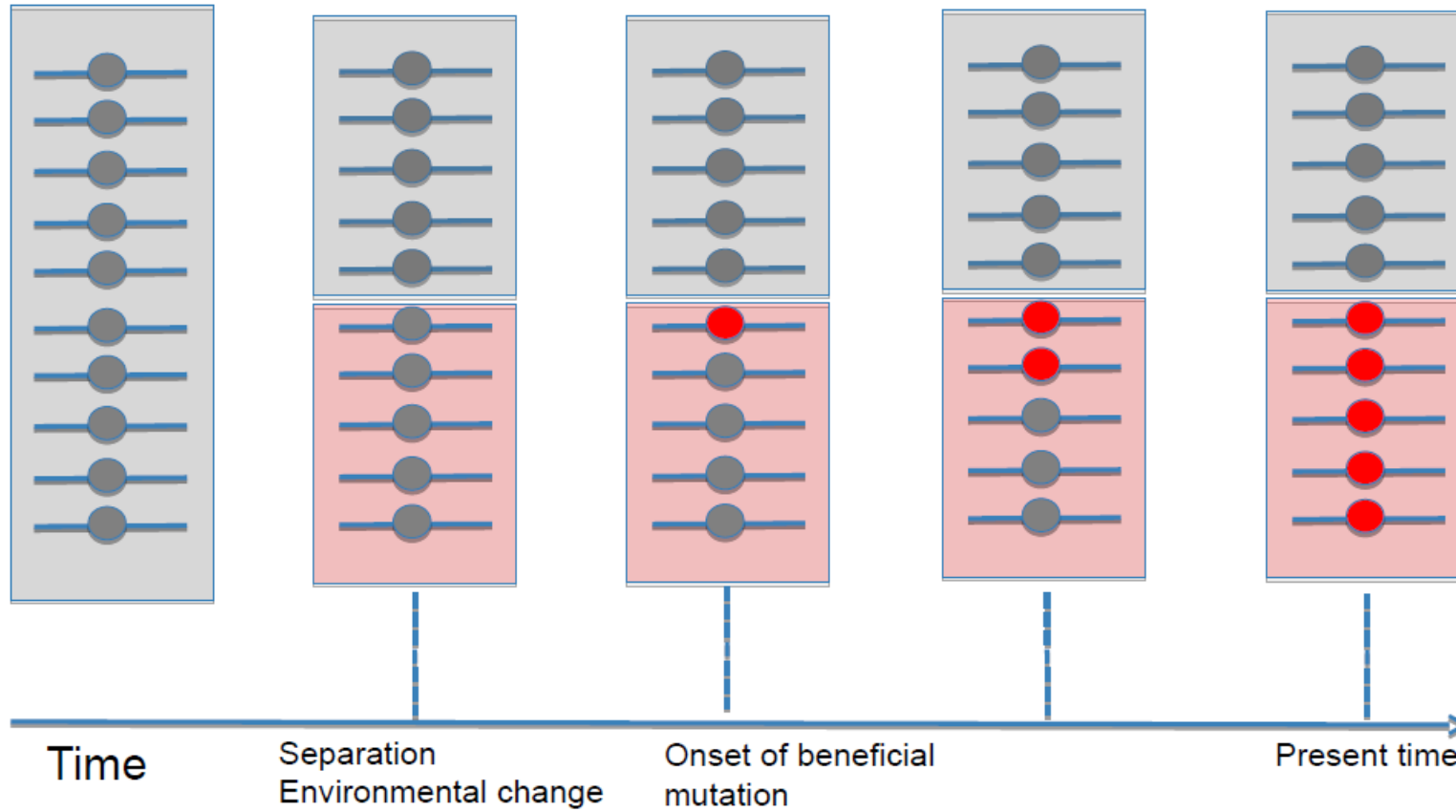
Allele Frequency Differentiation



Allele Frequency Differentiation

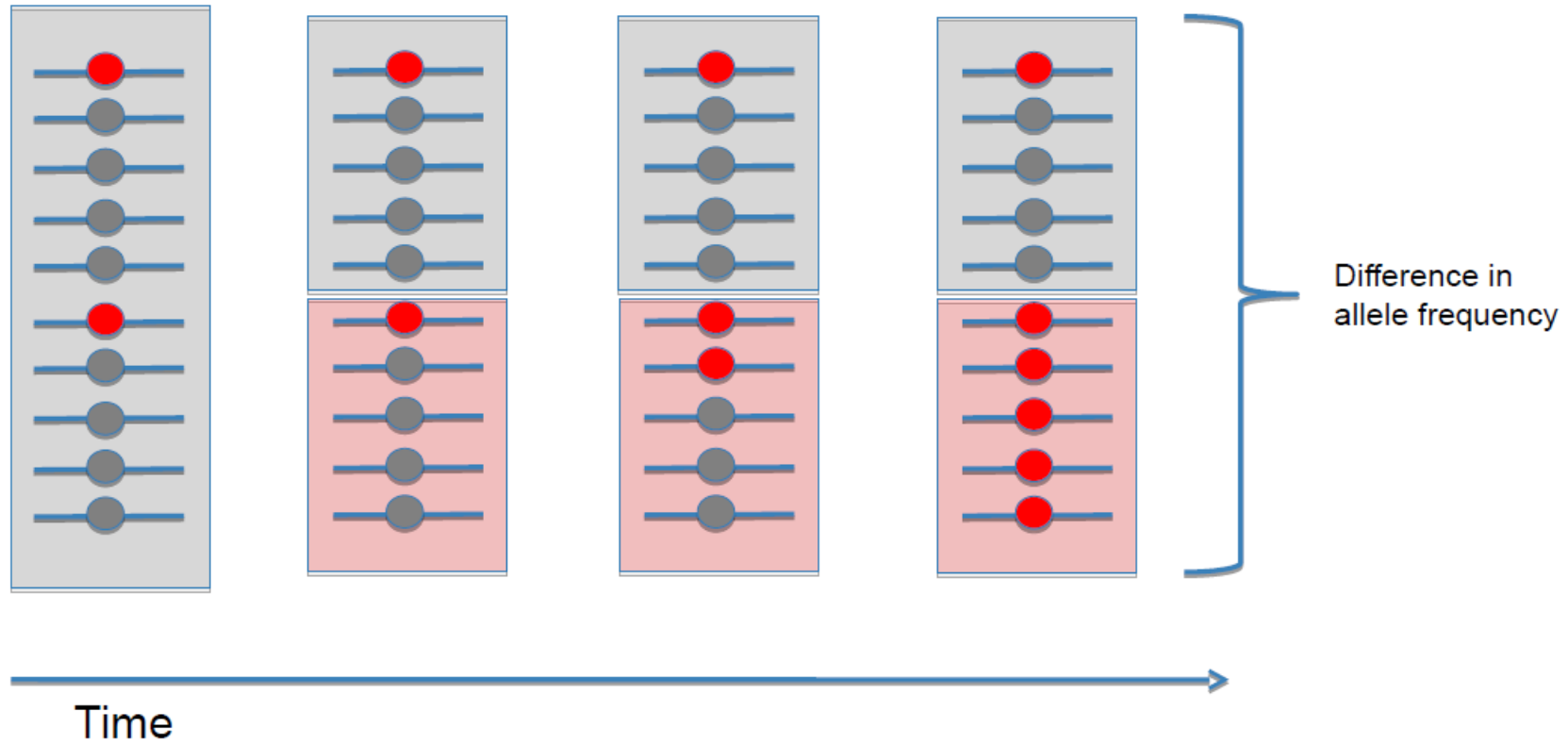


Allele Frequency Differentiation



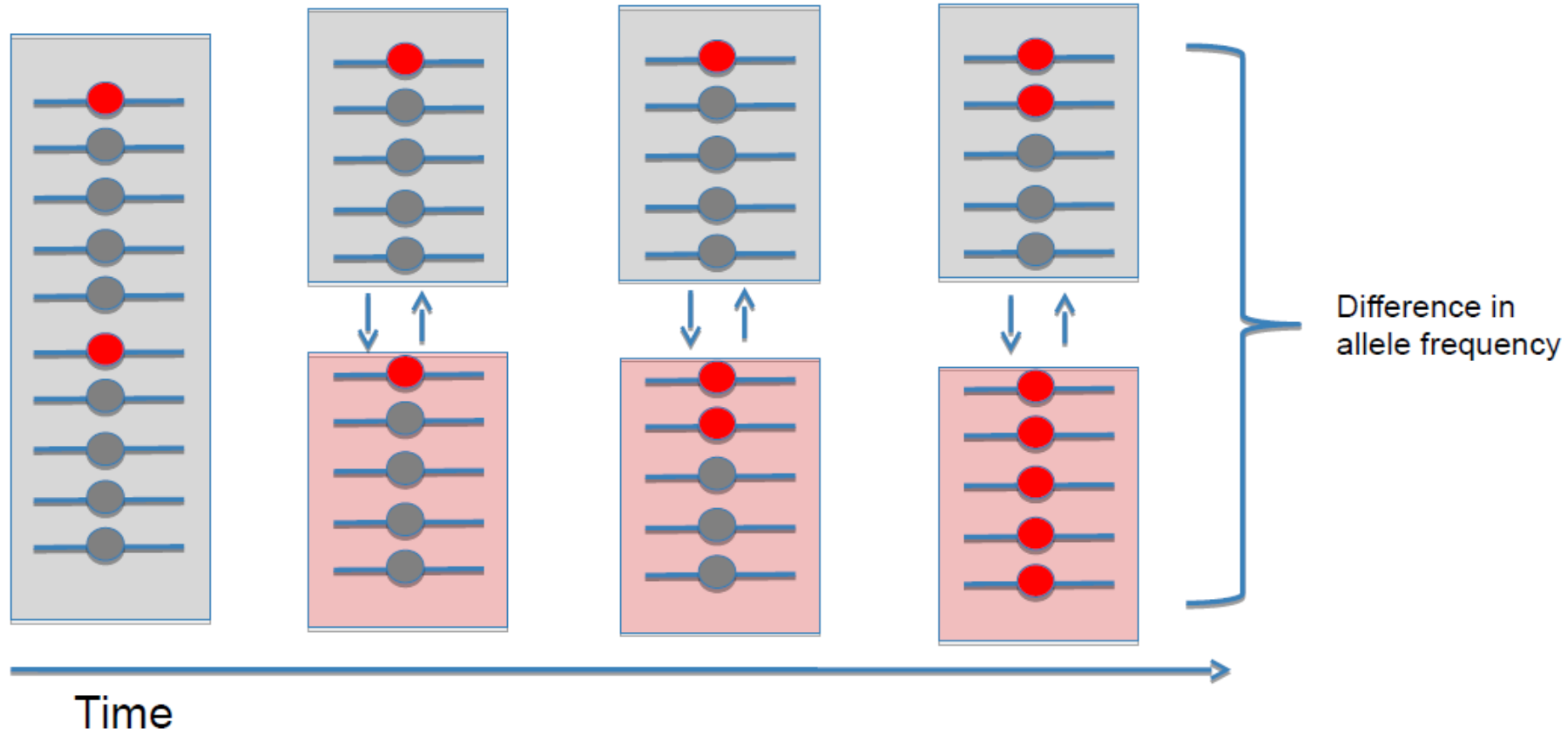
Allele Frequency Differentiation

From standing variation



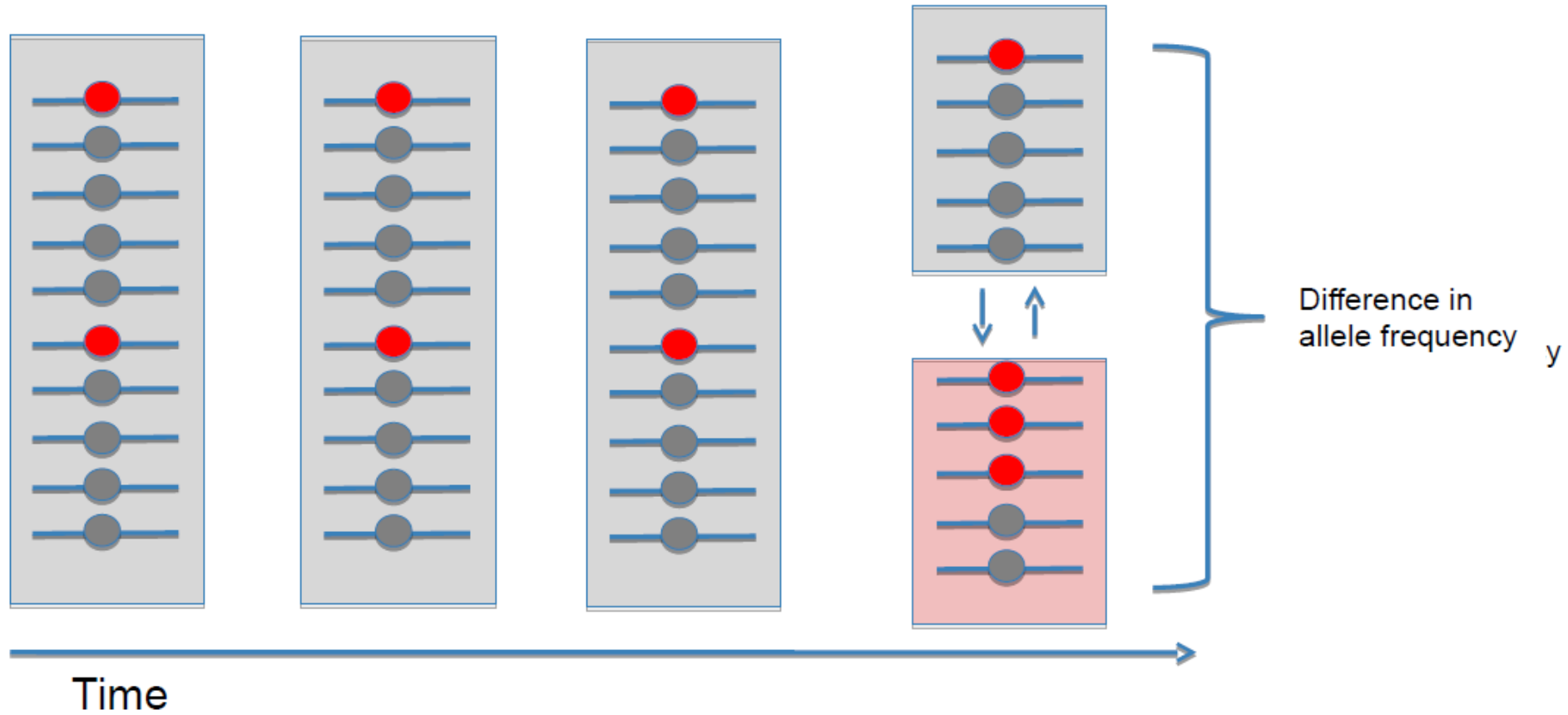
Allele Frequency Differentiation

With migration

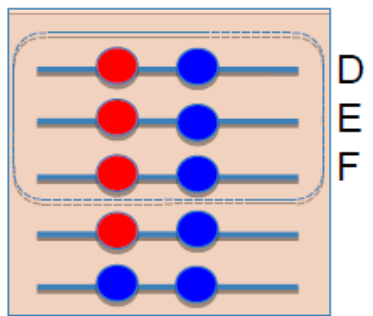
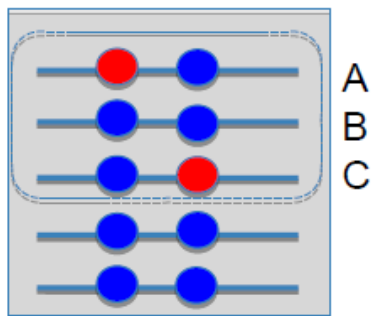


Allele Frequency Differentiation

With recent divergence



F_{ST} based on haplotype differentiation between populations



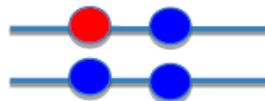
$$F_{ST} = 1 - (H_W / H_B)$$

Within populations

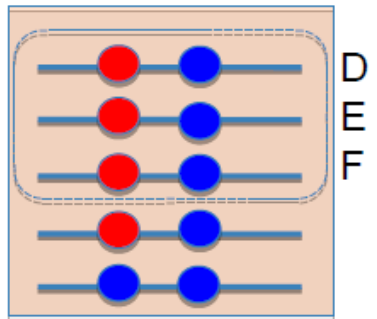
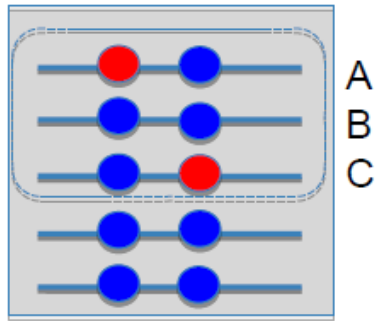
Between populations

What is the variation within populations?

e.g. A vs B



The differ by 1 site



$$F_{ST} = 1 - (H_W / H_B)$$

Within populations

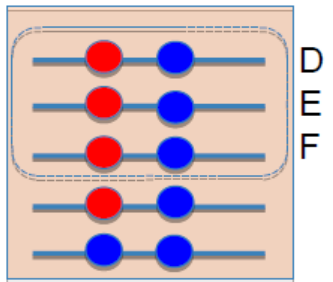
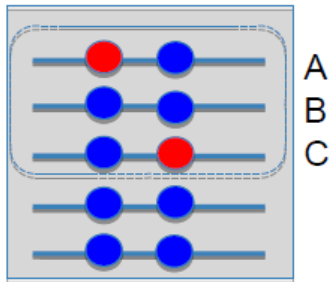
Between populations

What is the variation within populations?

A	B	Mean=?
A	C	
B	C	

D	E	Mean=?
D	F	
E	F	

H_W is the average within-populations: ?



$$F_{ST} = 1 - (H_W / H_B)$$

Within populations

Between populations

What is the variation within populations?

A	B	1	Mean=4/3
A	C	2	
B	C	1	

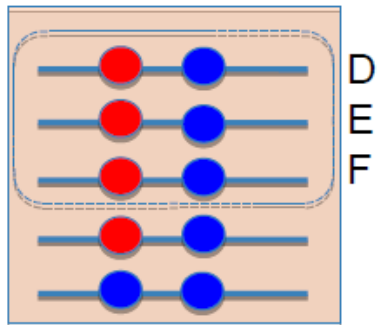
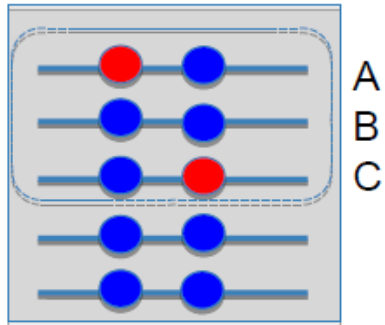
D	E	0	Mean=0/3
D	F	0	
E	F	0	

H_W is the average within-populations: $(4/3 + 0/3) / 2 = 2/3$

$$F_{ST} = 1 - (H_W / H_B)$$

Within populations

Between populations



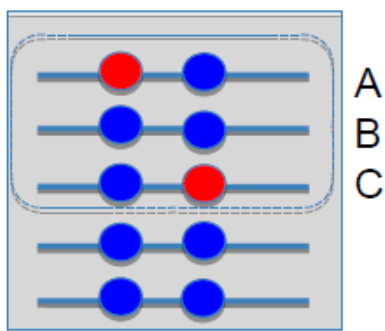
What is the variation between populations?

A	D	0
A	E	0
A	F	0
B	D	1
B	E	1
B	F	1
C	D	2
C	E	2
C	F	2

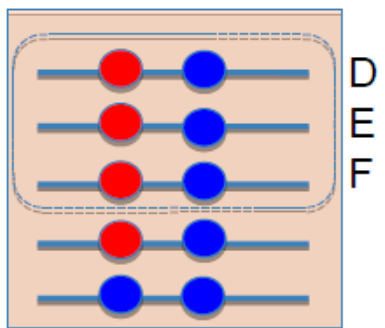
Mean=9/9

H_B is the average between-populations: $9/9=1$

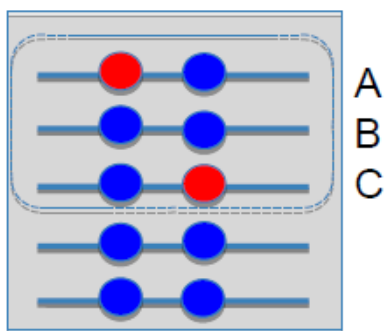
F_{ST} based on haplotype differentiation between populations



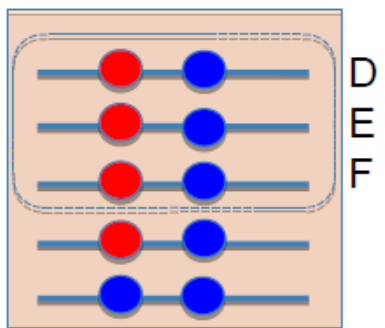
$$F_{ST} = 1 - (H_W / H_B) = 1 - ((2/3)/1) = 1/3 \sim 0.33$$



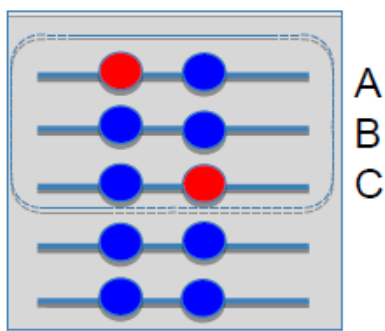
F_{ST} based on haplotype differentiation between populations



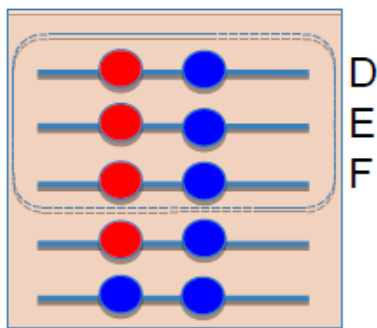
$$F_{ST} = 1 - (H_W / H_B) = 1 - ((2/3)/1) = 1/3 \sim 0.33$$



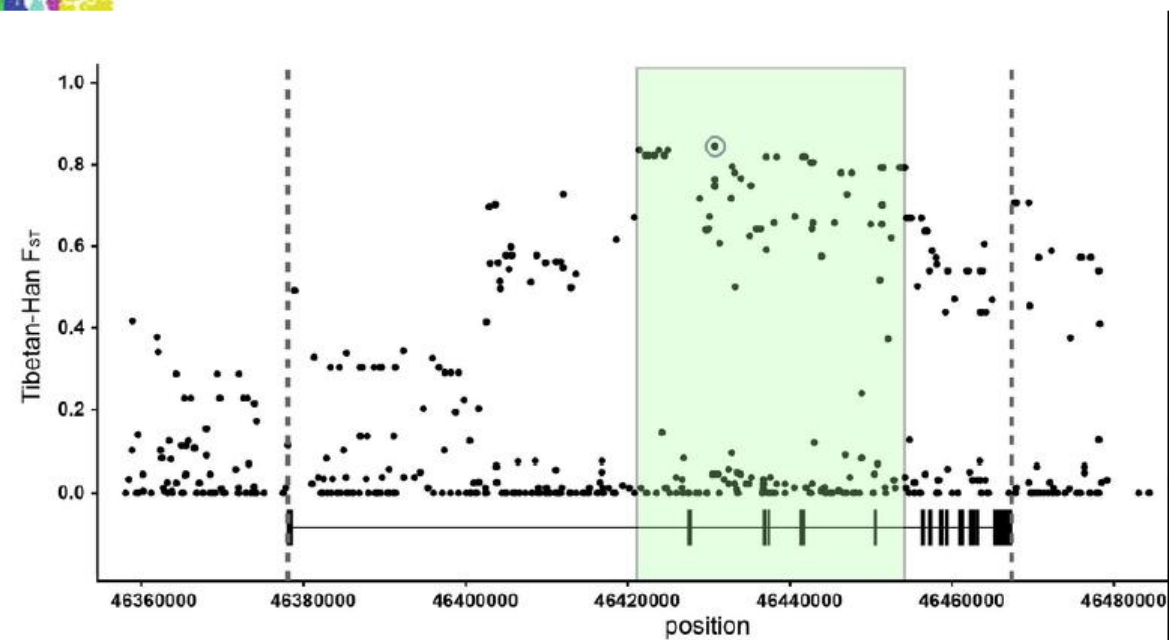
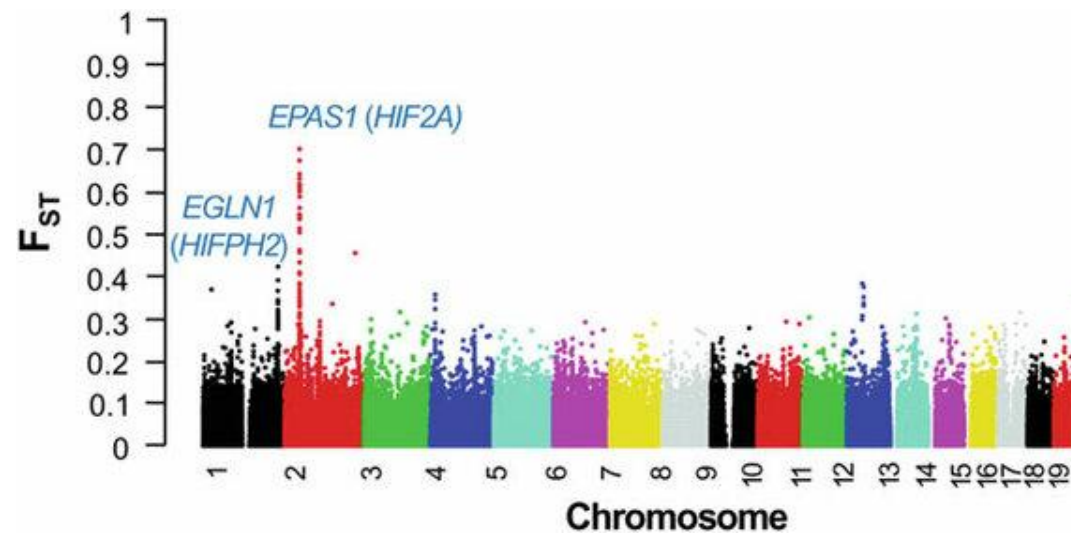
F_{ST} based on haplotype differentiation between populations



$$F_{ST} = 1 - (H_W / H_B) = 1 - ((2/3)/1) = 1/3 \sim 0.33$$

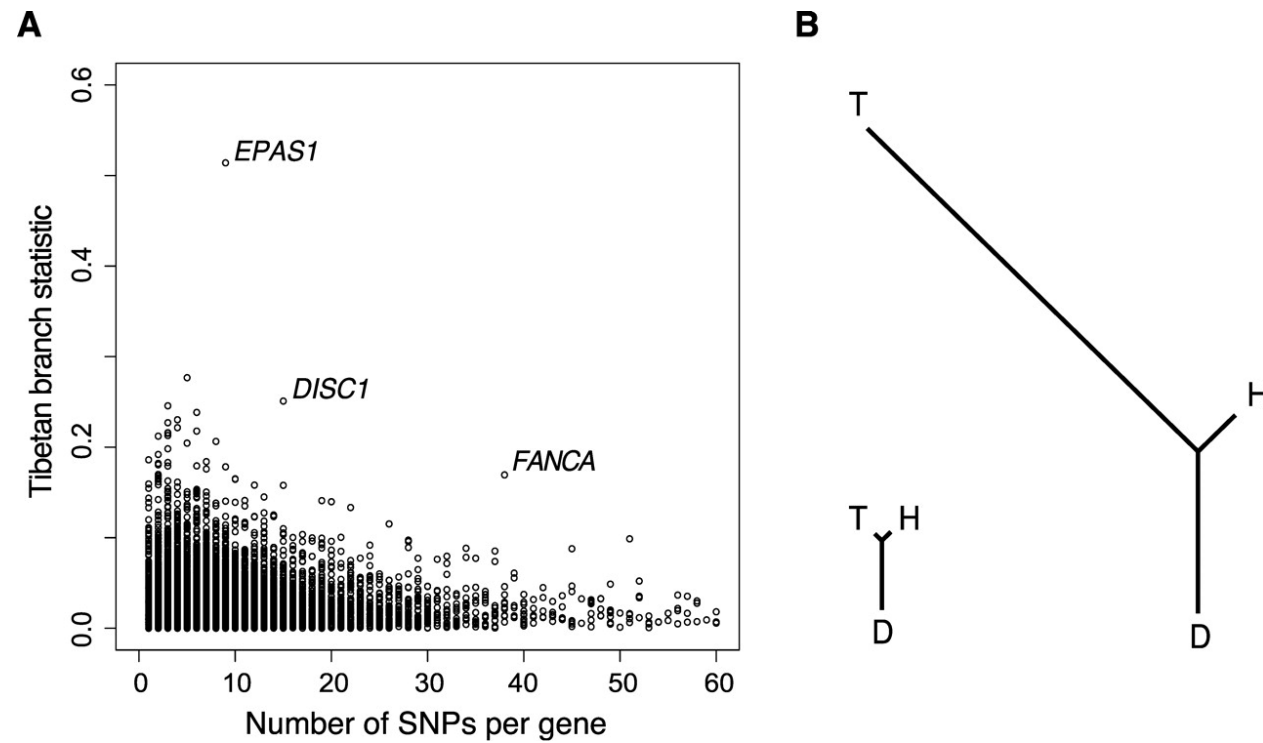


F_{ST}



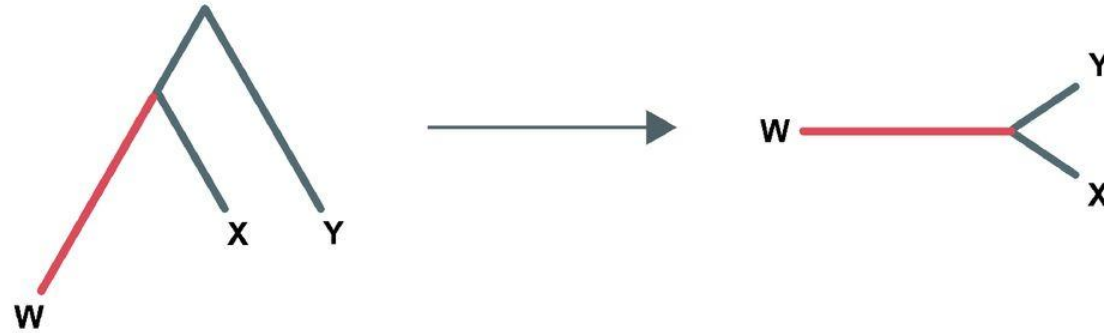
Allele	Tibetan	Han	Danish
C	0.13	0.9125	1
G	0.87	0.0875	0

Allele	Tibetan	Han	Danish
C	0.13	0.9125	1
G	0.87	0.0875	0



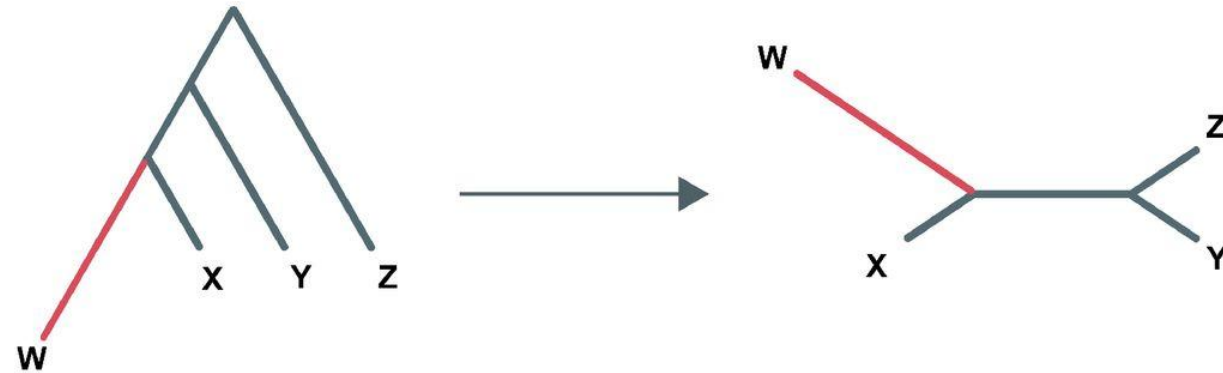
Population Branch Statistic (PBS)

A



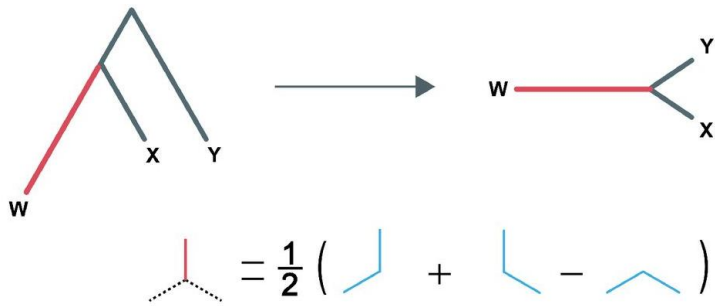
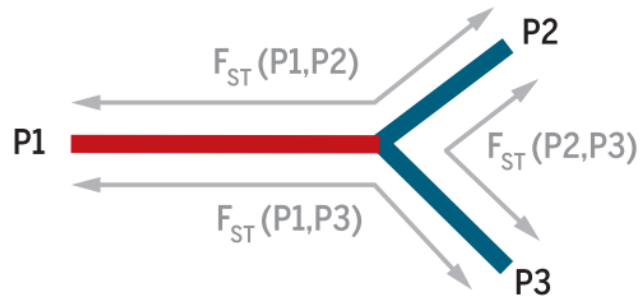
$$\text{[Tree Diagram]} = \frac{1}{2} \left(\text{[Tree Diagram 1]} + \text{[Tree Diagram 2]} - \text{[Tree Diagram 3]} \right)$$

B



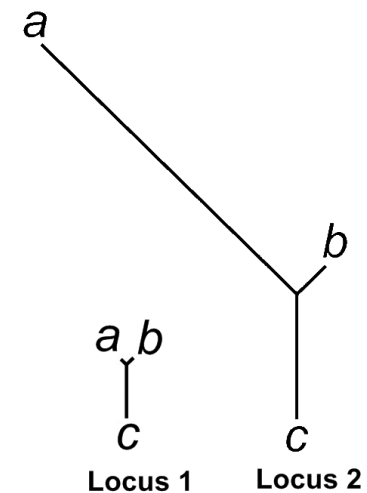
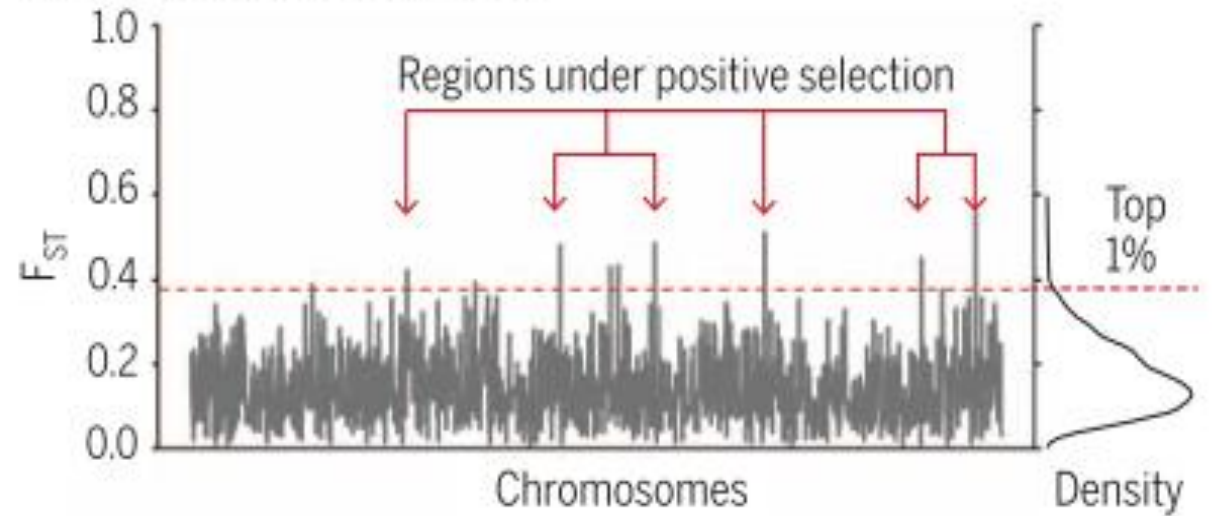
$$\text{[Tree Diagram]} = \frac{1}{4} \left(2 \text{[Tree Diagram 1]} + \text{[Tree Diagram 2]} + \text{[Tree Diagram 3]} - \text{[Tree Diagram 4]} - \text{[Tree Diagram 5]} \right)$$

Population Branch Statistic (PBS)



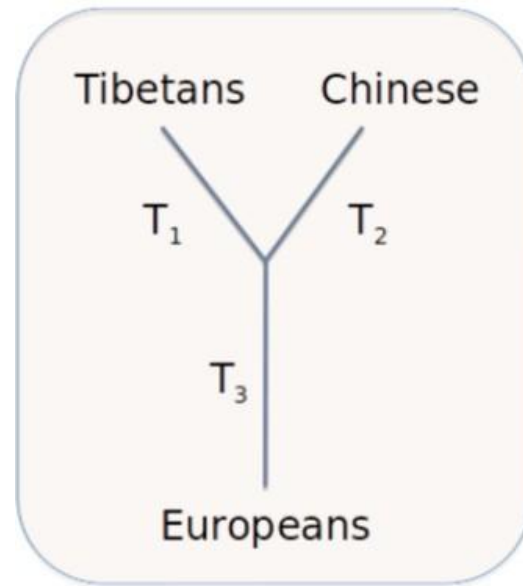
$$PBS = \frac{F_{ST}(P1;P2) + F_{ST}(P1;P3) - F_{ST}(P2;P3)}{2}$$

ii) Change in F_{ST} along genome

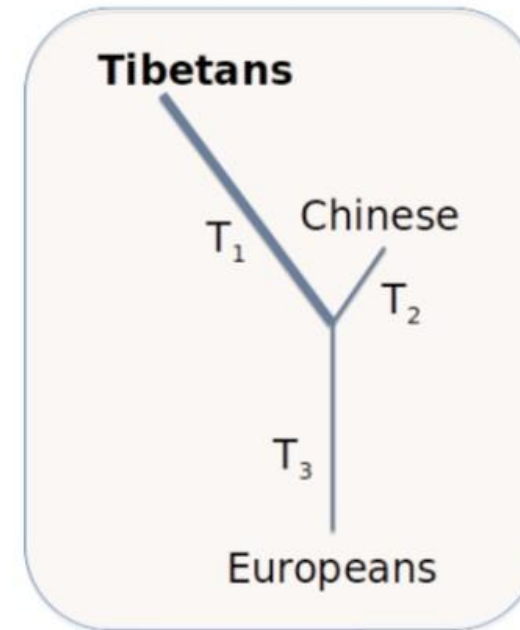


Population Branch Statistic (PBS)

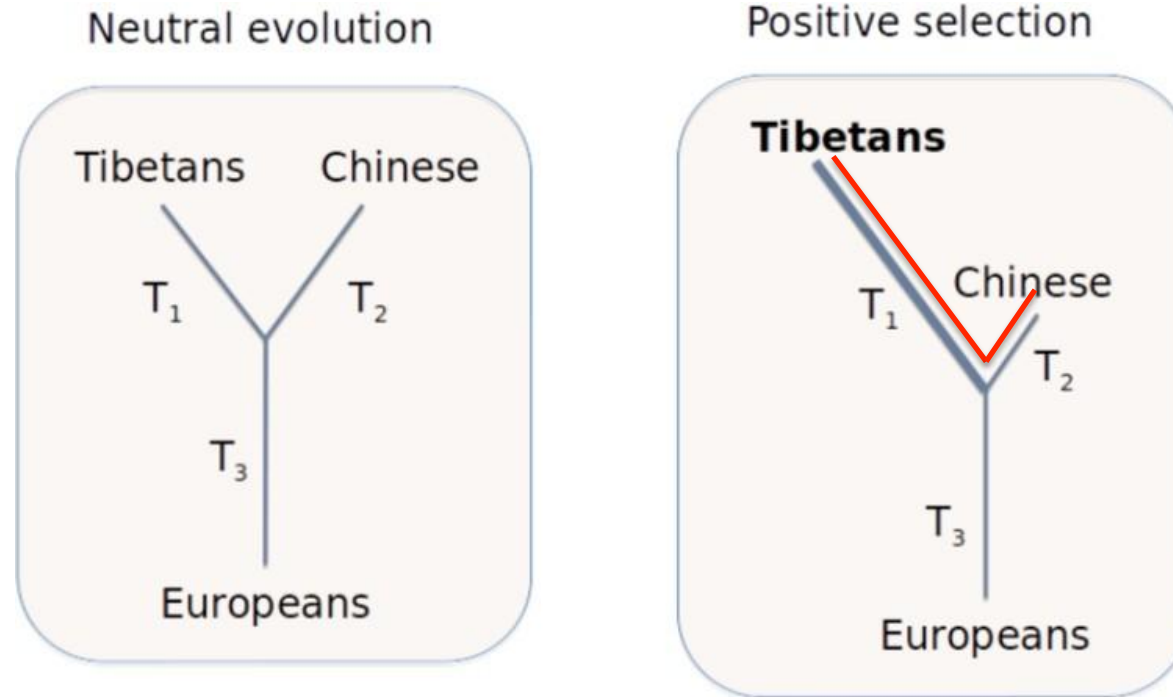
Neutral evolution



Positive selection

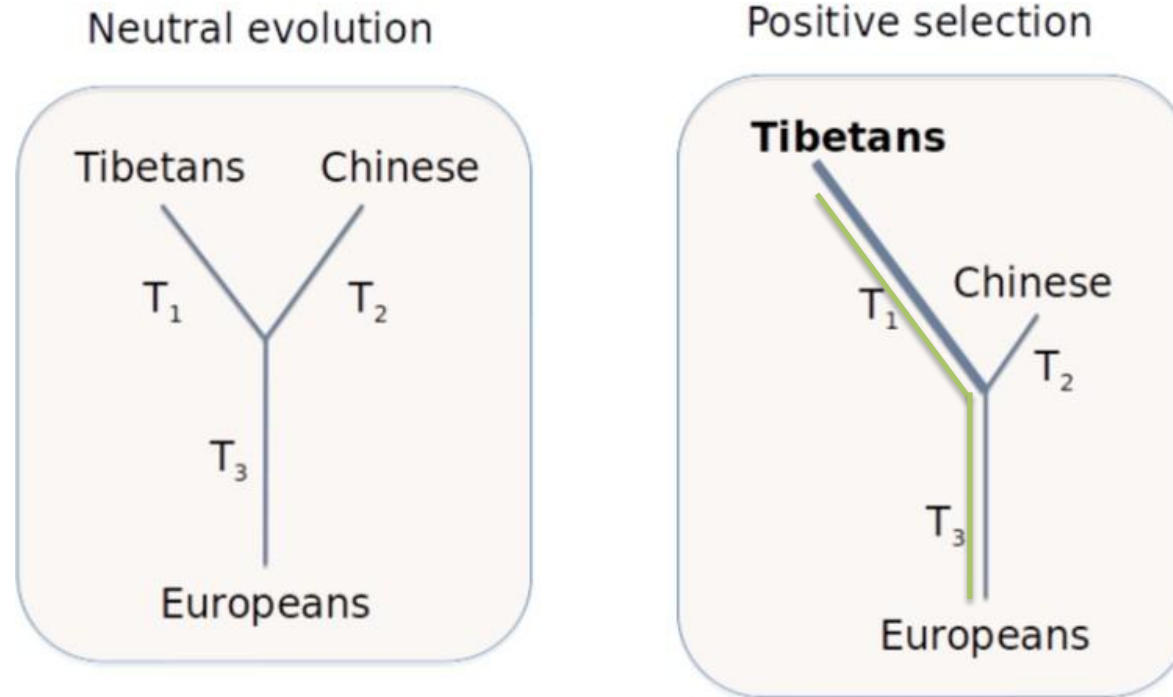


Population Branch Statistic (PBS)



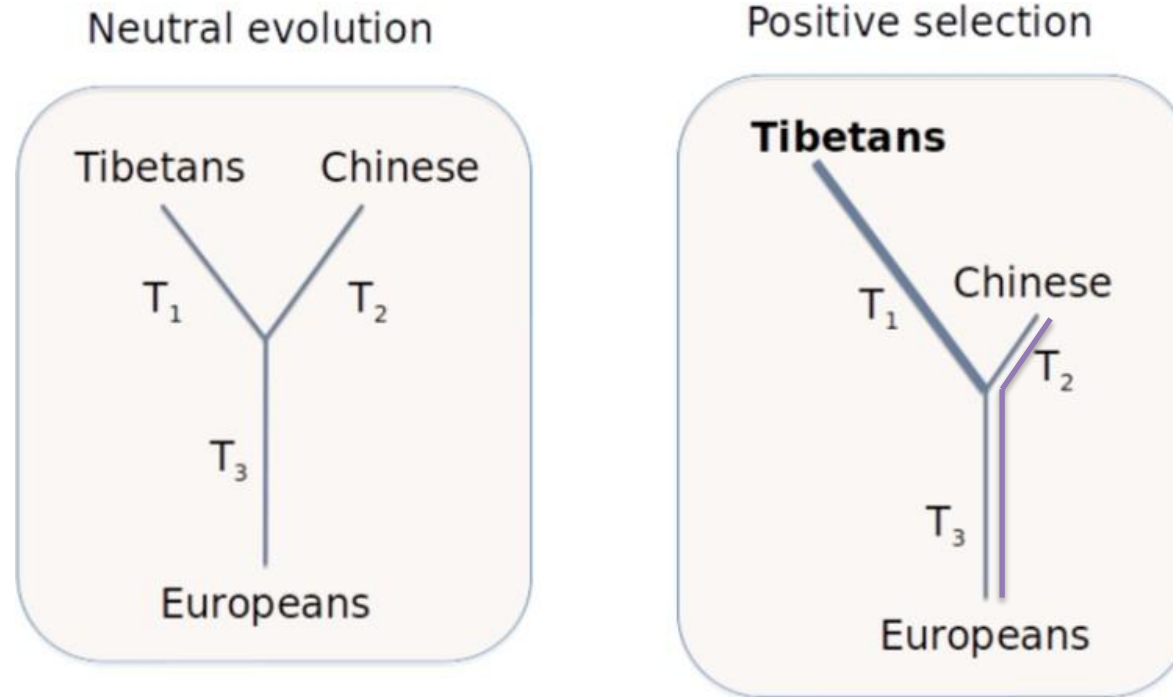
$$PBS = T_1T_2 + T_1T_3 - T_2T_3$$

Population Branch Statistic (PBS)



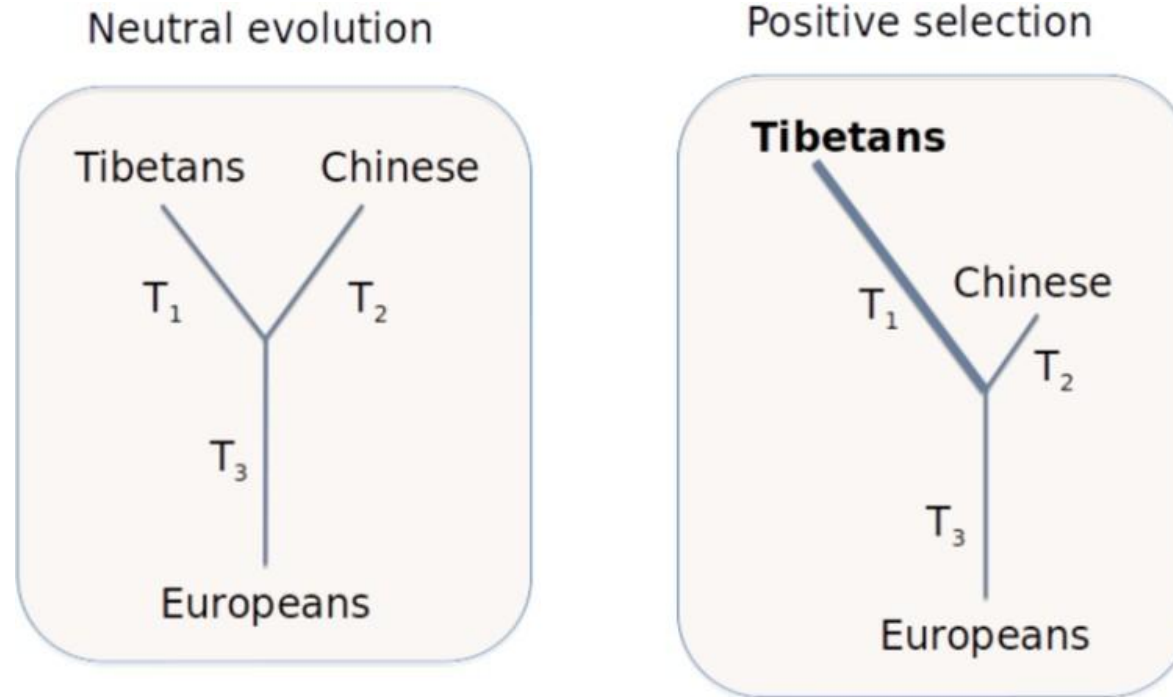
$$\text{PBS} = T_1T_2 + T_1T_3 - T_2T_3$$

Population Branch Statistic (PBS)



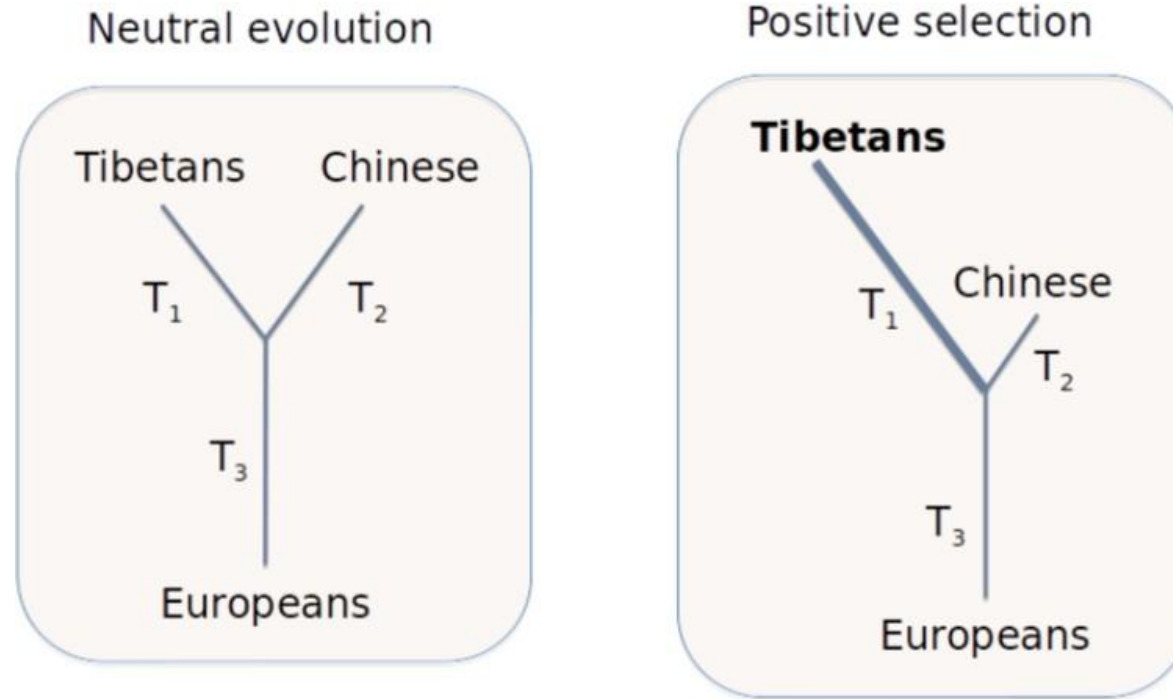
$$\text{PBS} = T_1T_2 + T_1T_3 - T_2T_3$$

Population Branch Statistic (PBS)



$$PBS = \frac{T_1T_2 + T_1T_3 - T_2T_3}{2}$$

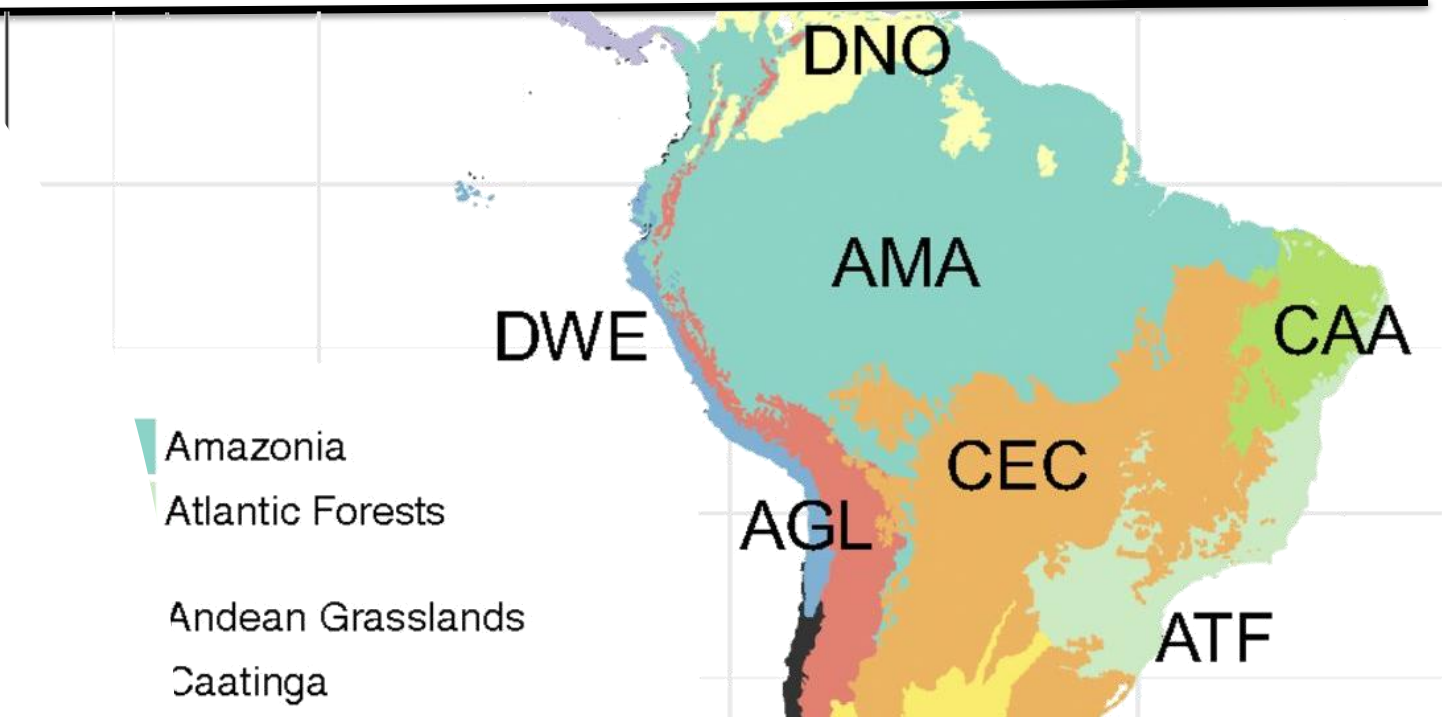
Population Branch Statistic (PBS)



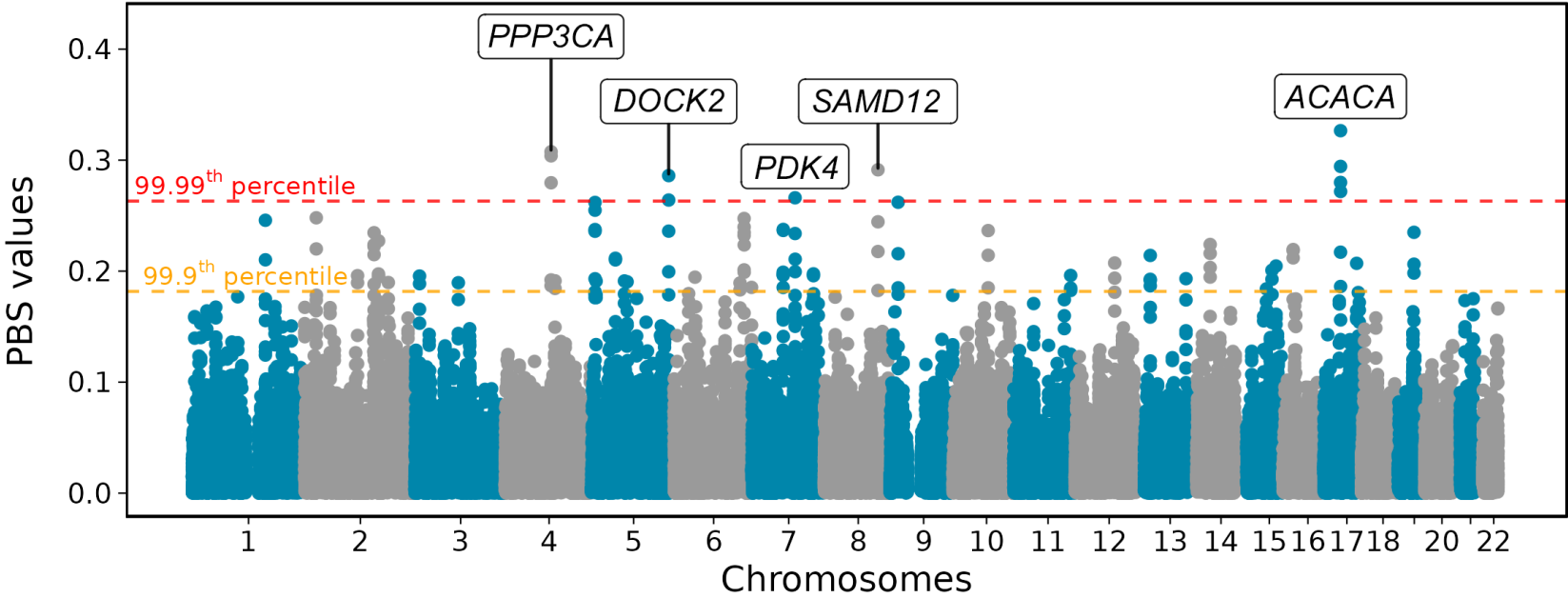
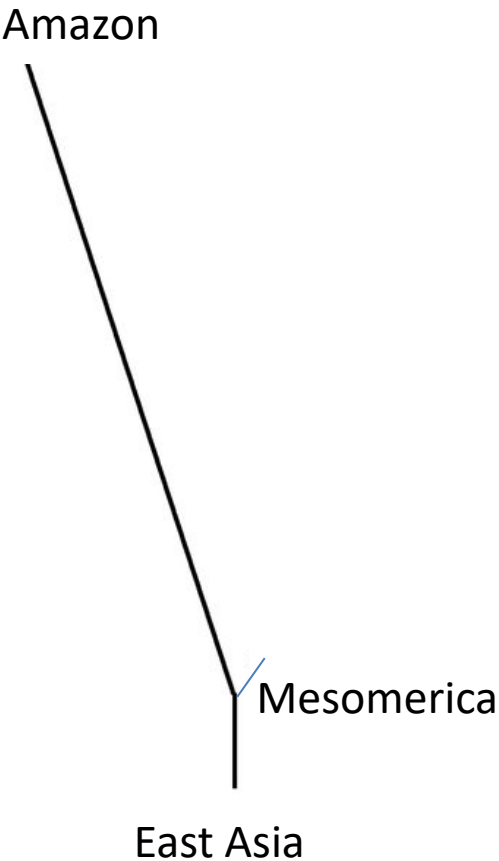
$$PBS = F_{ST_TIB_CHB} + F_{ST_TIB_EUR} - F_{ST_CHB_EUR} / 2$$

Population Branch Statistic (PBS)

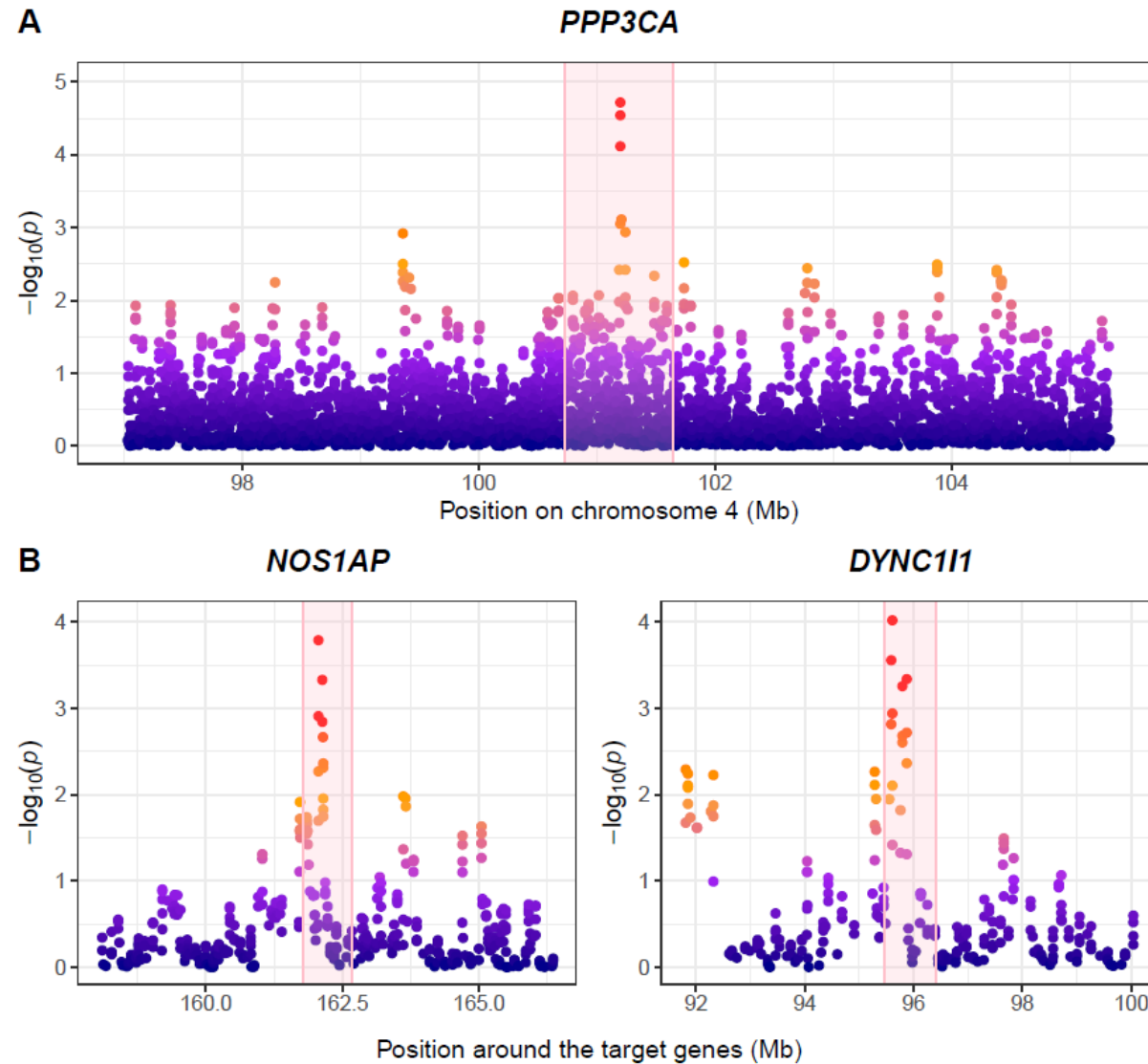
- Higher plant diversity;
- Lower UV radiation;
- Food source instability;
- Highest pathogen presence (70%)
- Shorter lifespan;
- Thermoregulation;



Population Branch Statistic (PBS)



Population Branch Statistic (PBS)



Population Branch Statistic (PBS)

- **PBS (Population Branch Statistic):** Measures allele frequency differentiation along a focal population's branch versus two reference populations, providing branch length analogues via F_{ST} -based genetic divergence.
- **PBSn1 (Normalized PBS):** Further rescales PBS by accounting for the branch lengths in the outgroups at the same locus.
- **PBE (Population Branch Excess):** Adjusts PBSn1 using the genome-wide median divergence, making it more robust to background selection and parallel sweeps.

Alternative? PCAdapt

Robustness: Handles complex admixture and cryptic population structure better than FST outlier approaches.

Efficiency: Fast computation times on massive, multi-gigabyte genomic datasets.

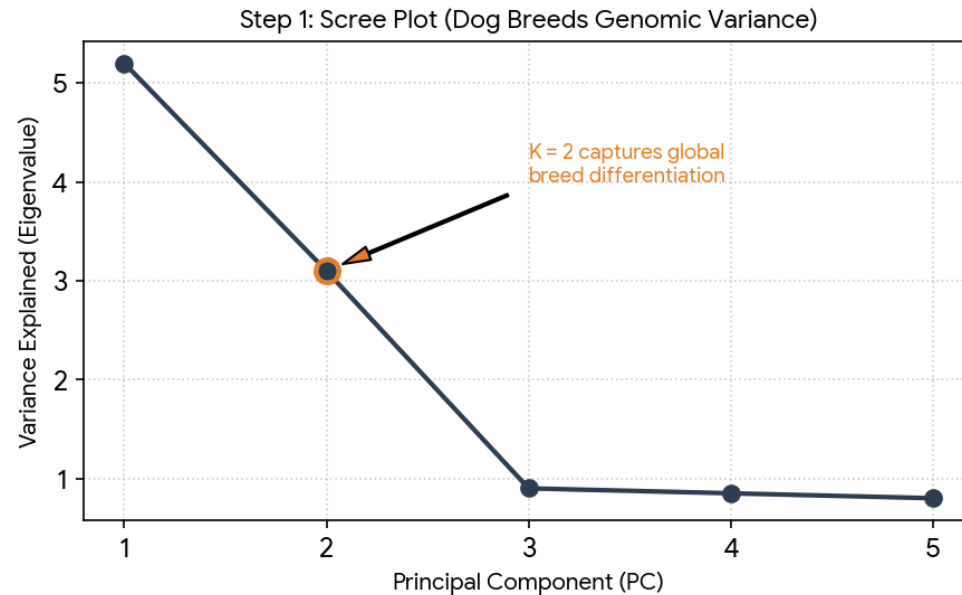
Check: Essential to follow up outliers with functional annotation or gene ontology analysis to validate biological adaptation.

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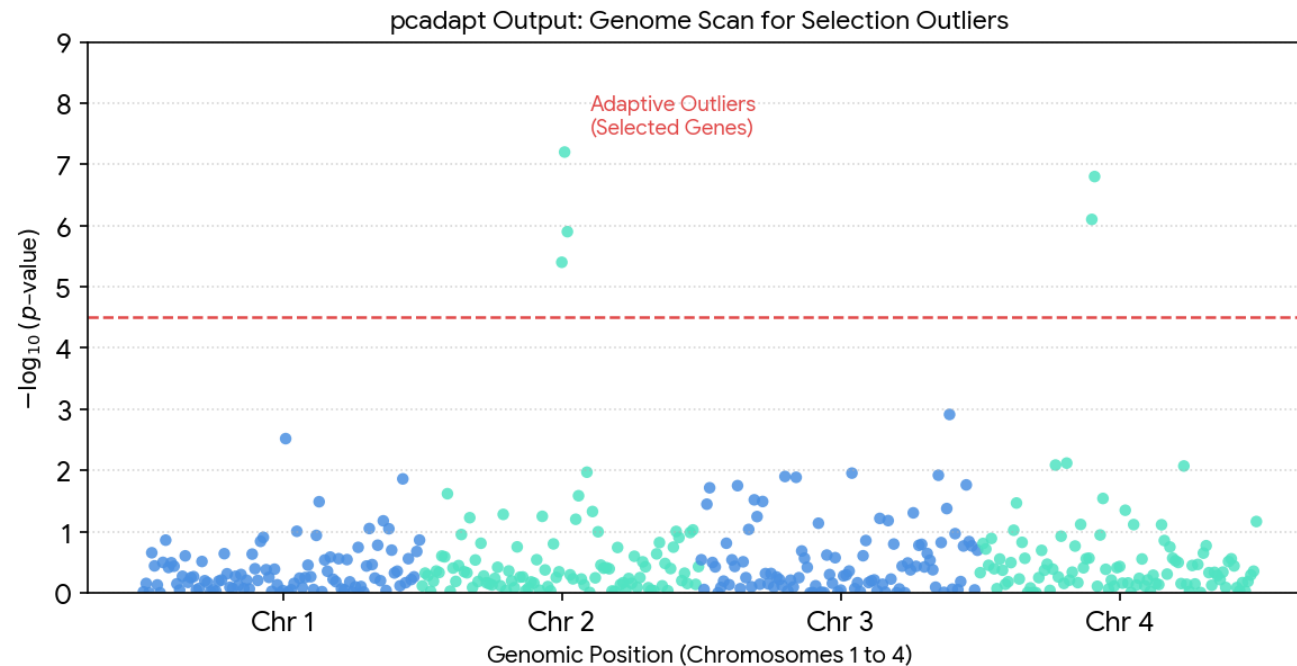


Alternative? PCAdapt

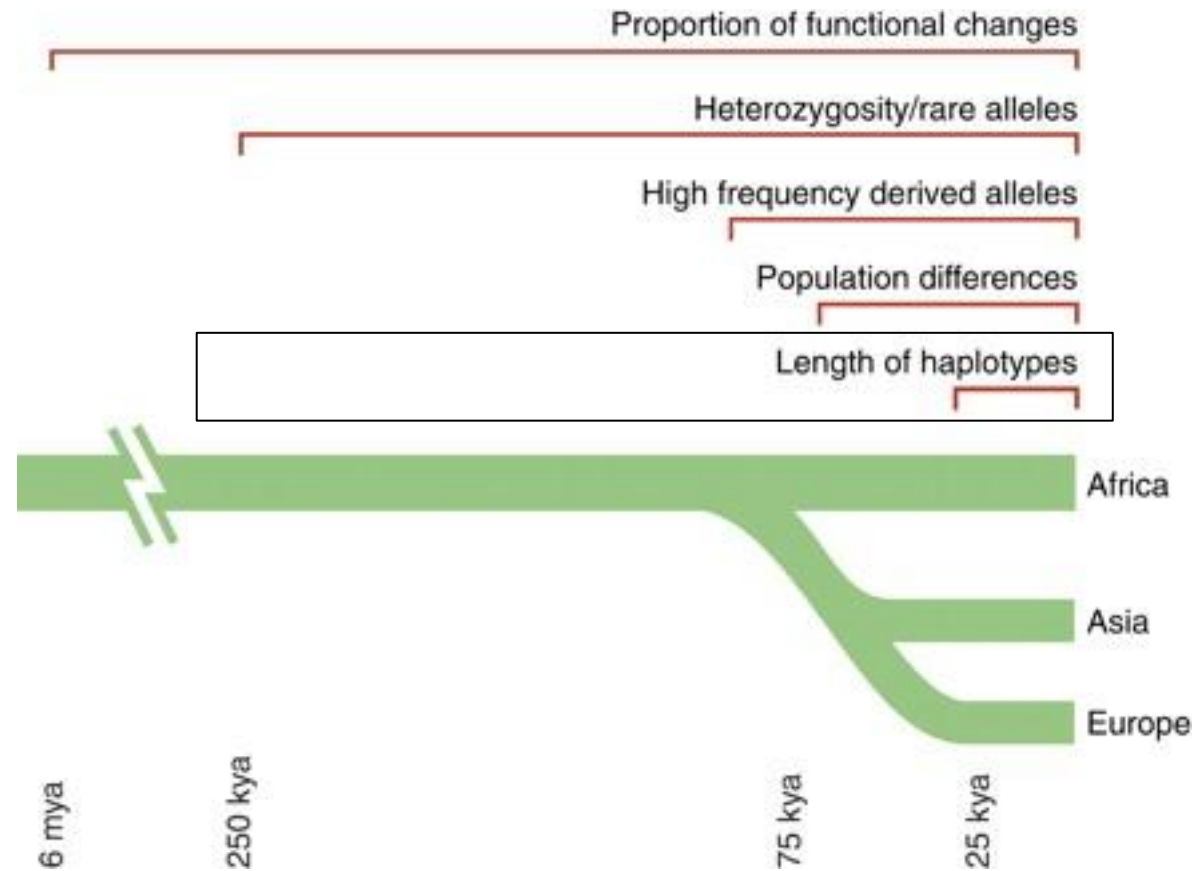
Z-scores: Quantify how much each SNP correlates with PC1, PC2, ..., PC_K.

Robust Covariance Matrix: Evaluates the background distribution by ignoring potential outliers.

Mahalanobis Distance: Measures how far a locus is from the center of the neutral genomic cloud.



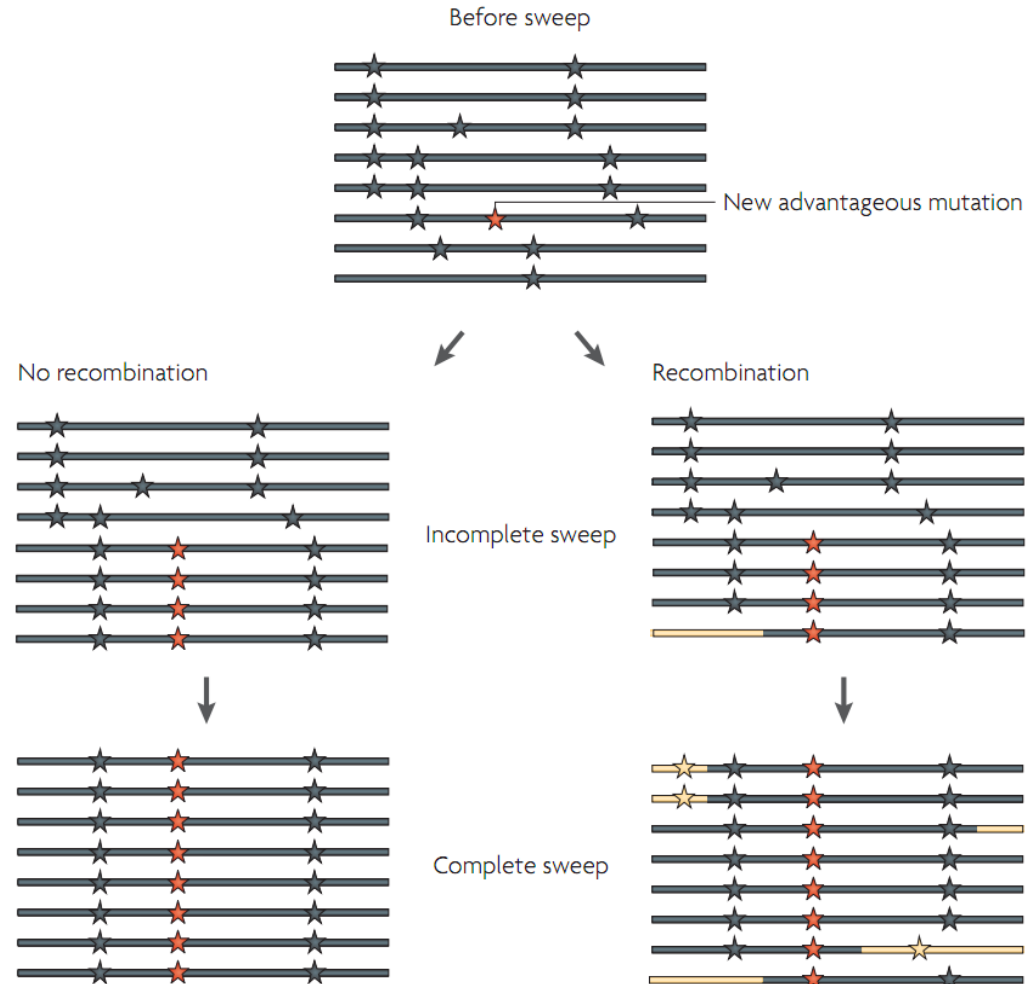
Testes de Seleção intra-específicos



FONTE: Sabeti PC, et al. Positive natural selection in the human lineage. Science. 2006 Jun 16;312(5780):1614-20. Review.

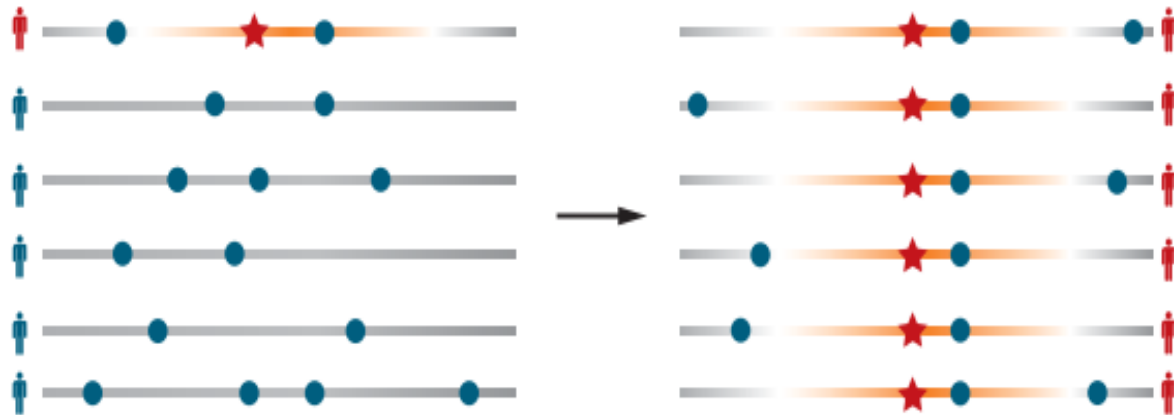
Extended Haplotype Homozygosity (EHH)

Extended haplotype homozygosity (EHH): EHH at distance x from the core region is the probability that two randomly chosen chromosomes carry a tested core haplotype are homozygous at all SNPs for the entire interval from the core region to the distance x .



Extended Haplotype Homozygosity (EHH)

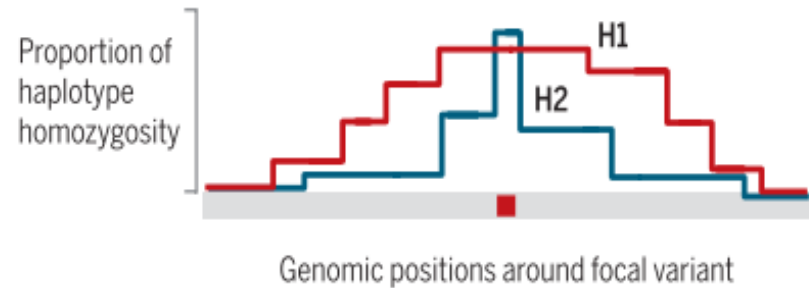
Hard sweep



iv) Haplotype homozygosity between two haplotypes (H1, H2)



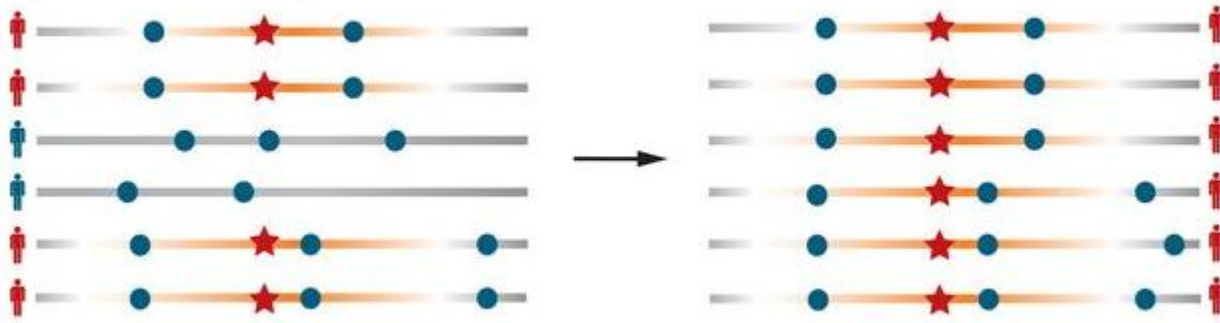
iv) Extended haplotype homozygosity (EHH)



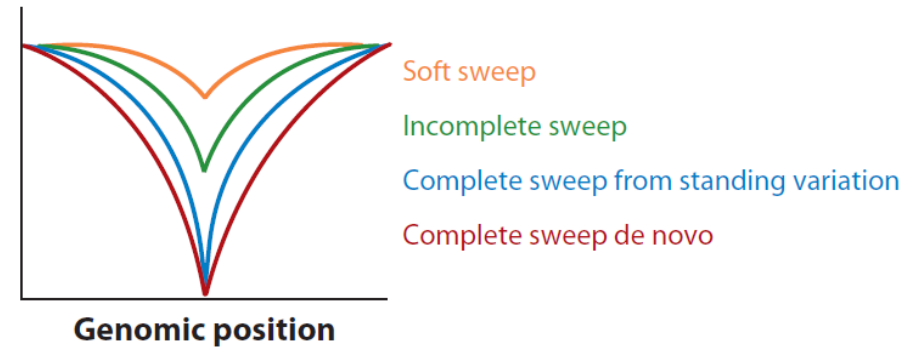
Extended Haplotype Homozygosity (EHH)

Soft sweep

ii) Selection on standing variations



a Population diversity



Extended Haplotype Homozygosity (EHH)

	1	
	1	
	1	
	1	
	1	
	0	



Core allele 1 : biallelic loci; 0 is the ancestral allele and 1 is the derived allele.

Extended Haplotype Homozygosity (EHH)

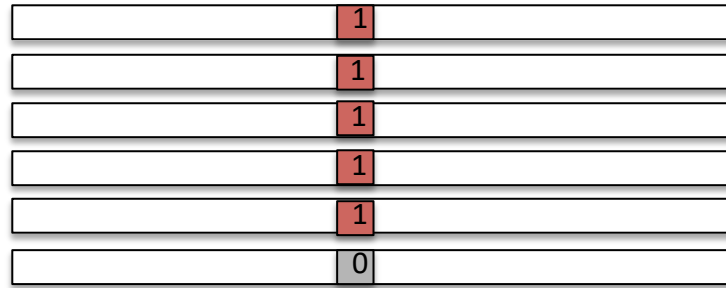
	1	
	1	
	1	
	1	
	1	
	0	



Core allele 1 : biallelic loci; 0 is the ancestral allele and 1 is the derived allele.

$$EHH_c(x_i) = \sum_{h \in H_c(x_i)} \frac{\binom{n_h}{2}}{\binom{n_c}{2}}$$

Extended Haplotype Homozygosity (EHH)

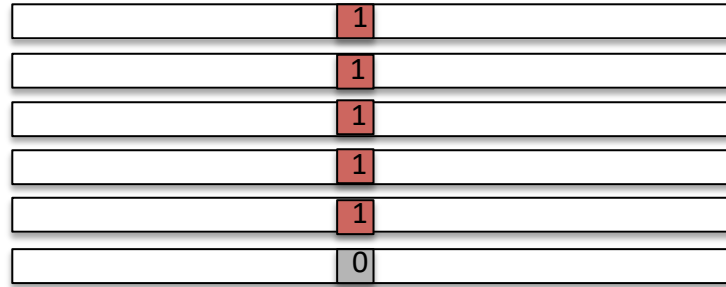


Core allele 1 : biallelic loci; 0 is the ancestral allele and 1 is the derived allele.

$$EHH_c(x_i) = \sum_{h \in H_c(x_i)} \frac{\binom{n_h}{2}}{\binom{n_c}{2}}$$

From x_0 to x_i

Extended Haplotype Homozygosity (EHH)



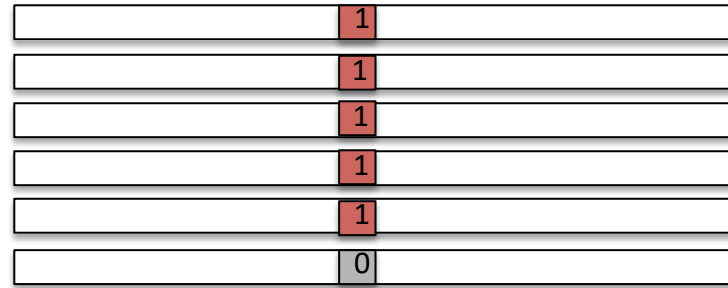
Core allele 1 : biallelic loci; 0 is the ancestral allele and 1 is the derived allele.

$$EHH_c(x_i) = \sum_{h \in H_c(x_i)} \frac{\binom{n_h}{2}}{\binom{n_c}{2}}$$

An arrow points from the summation symbol in the equation to the text below.

Sum of all haplotypes containing the allele of interest (core allele)

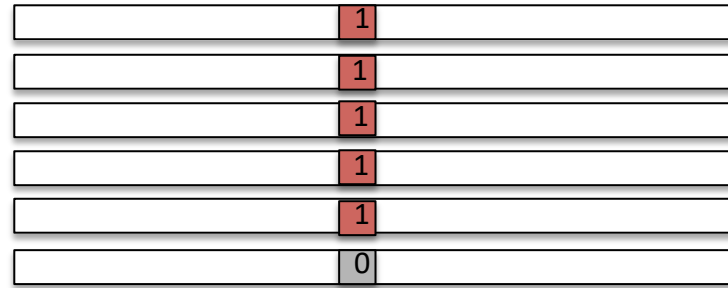
Extended Haplotype Homozygosity (EHH)



Core allele 1 : biallelic loci; 0 is the ancestral allele and 1 is the derived allele.

$$EHH_c(x_i) = \sum_{h \in H_c(x_i)} \frac{\binom{n_h}{2}}{\binom{n_c}{2}} \quad n_h \text{ haplotype Frequency of } h$$

Extended Haplotype Homozygosity (EHH)



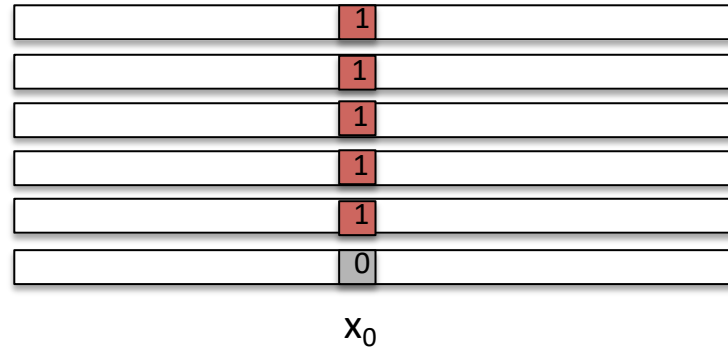
Core allele 1 : biallelic loci; 0 is the ancestral allele and 1 is the derived allele.

$$EHH_c(x_i) = \sum_{h \in H_c(x_i)} \frac{\binom{n_h}{2}}{\binom{n_c}{2}}$$

n_h haplotype Frequency of h

N_c haplotype Frequency with core SNP

Extended Haplotype Homozygosity (EHH)



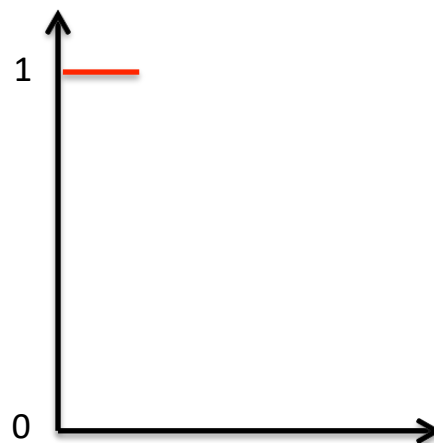
What is the EHH to x_0 ?

$$EHH_c(x_i) = \sum_{h \in H_c(x_i)} \frac{\binom{n_h}{2}}{\binom{n_c}{2}}$$
$$EHH_c(x_i = 0) = \frac{\binom{5}{2}}{\binom{5}{2}} = 1$$

Extended Haplotype Homozygosity (EHH)

	1
	1
	1
	1
	1
	0

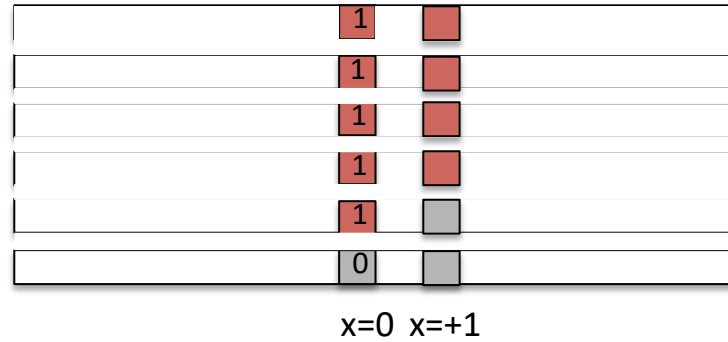
$x=0$



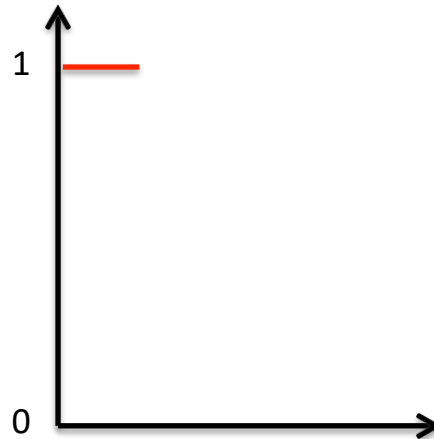
$$EHH_c(x_i = 0) = \frac{\binom{5}{2}}{\binom{5}{2}} = 1$$

$$EHH_c(x_i) = \sum_{h \in H_c(x_i)} \frac{\binom{n_h}{2}}{\binom{n_c}{2}}$$

Extended Haplotype Homozygosity (EHH)

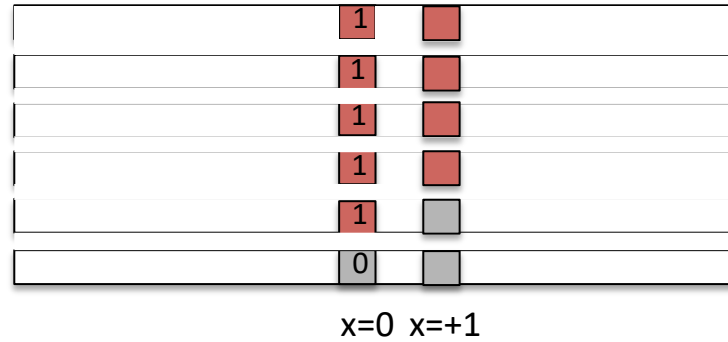


$$EHH_c(x_i) = \sum_{h \in H_c(x_i)} \frac{\binom{n_h}{2}}{\binom{n_c}{2}}$$

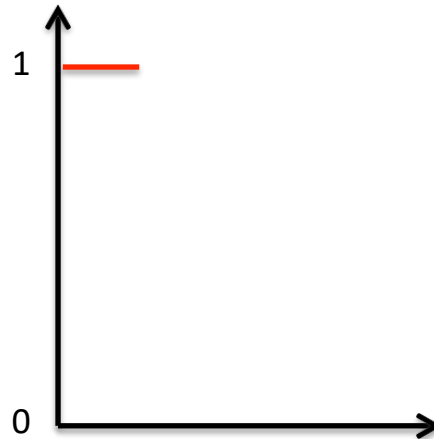


$$EHH_c(x_i = +1) = ?$$

Extended Haplotype Homozygosity (EHH)

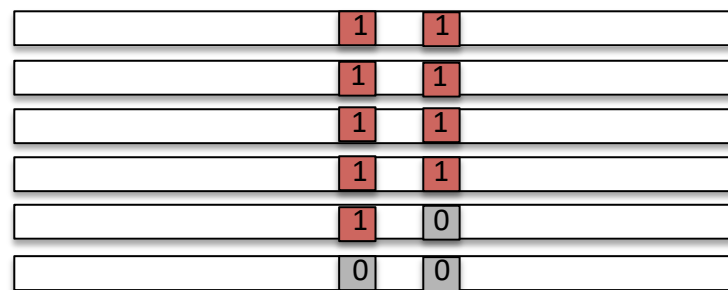


$$EHH_c(x_i) = \sum_{h \in H_c(x_i)} \frac{\binom{n_h}{2}}{\binom{n_c}{2}}$$

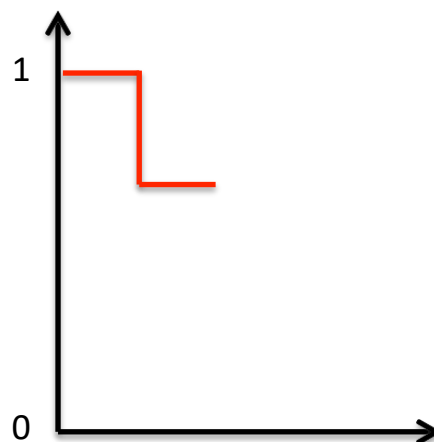


$$EHH_c(x_i = +1) = \frac{\binom{4}{2} + \binom{1}{2}}{\binom{5}{2}} = \frac{6 + 0}{10} = 0.60$$

Extended Haplotype Homozygosity (EHH)



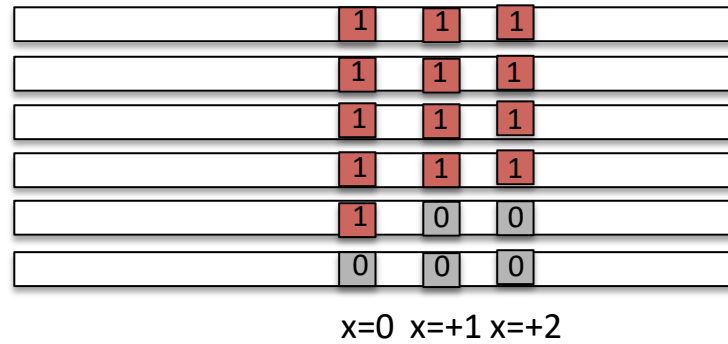
x=0 x=+1



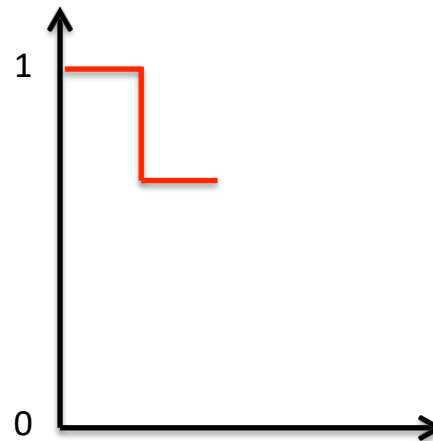
$$EHH_c(x_i) = \sum_{h \in H_c(x_i)} \frac{\binom{n_h}{2}}{\binom{n_c}{2}}$$

$$EHH_c(x_i = +1) = \frac{\binom{4}{2} + \binom{1}{2}}{\binom{5}{2}} = \frac{6 + 0}{10} = 0.60$$

Extended Haplotype Homozygosity (EHH)

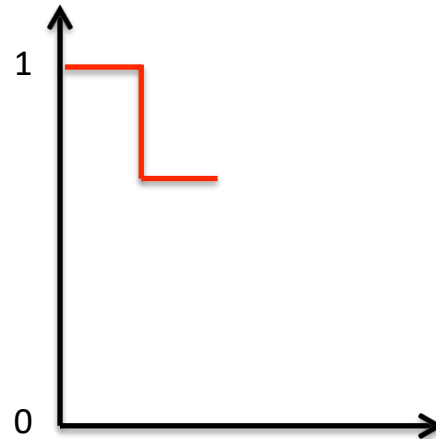
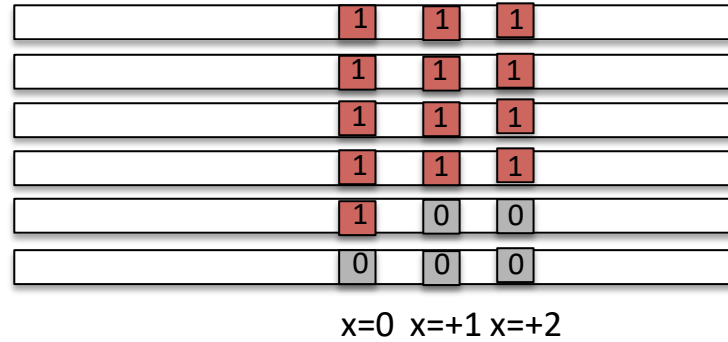


$$EHH_c(x_i) = \sum_{h \in H_c(x_i)} \frac{\binom{n_h}{2}}{\binom{n_c}{2}}$$



$$EHH_c(x_i = +2) = ?$$

Extended Haplotype Homozygosity (EHH)

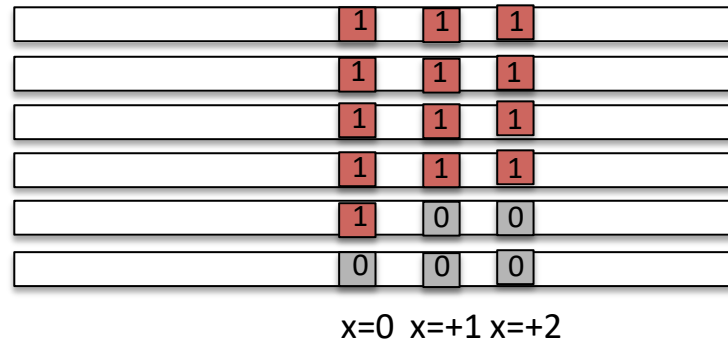


$$EHH_c(x_i) = \sum_{h \in H_c(x_i)} \frac{\binom{n_h}{2}}{\binom{n_c}{2}}$$

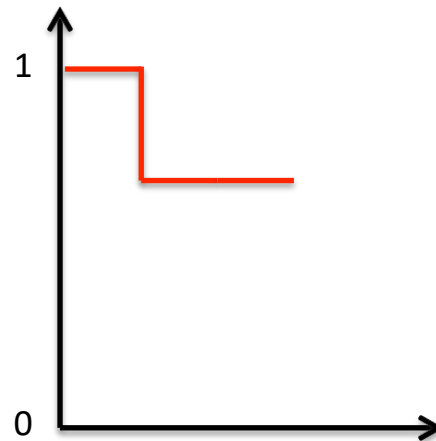
$$EHH_c(x_i = +2) = ?$$

How many unique haplotypes carrying the core SNP?
What is their frequency?

Extended Haplotype Homozygosity (EHH)

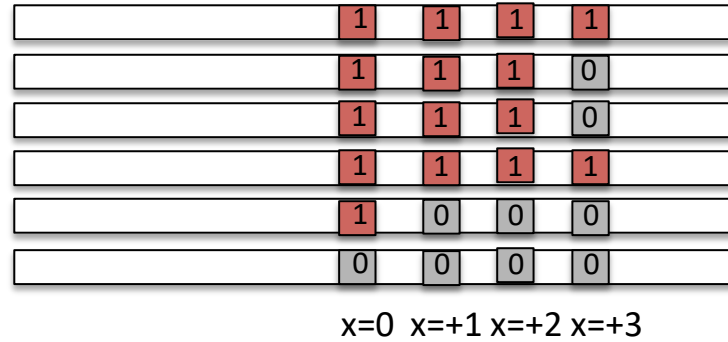


$$EHH_c(x_i) = \sum_{h \in H_c(x_i)} \frac{\binom{n_h}{2}}{\binom{n_c}{2}}$$

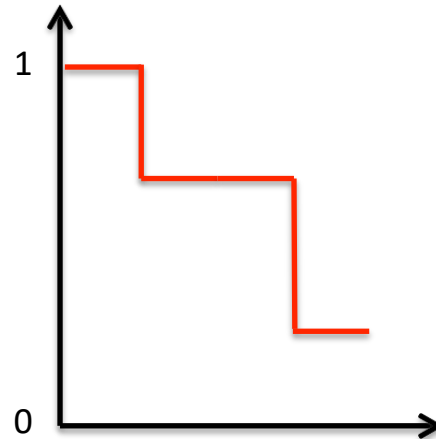


$$EHH_c(x_i = +2) = EHH_c(x_i = +1) = 0.60$$

Extended Haplotype Homozygosity (EHH)

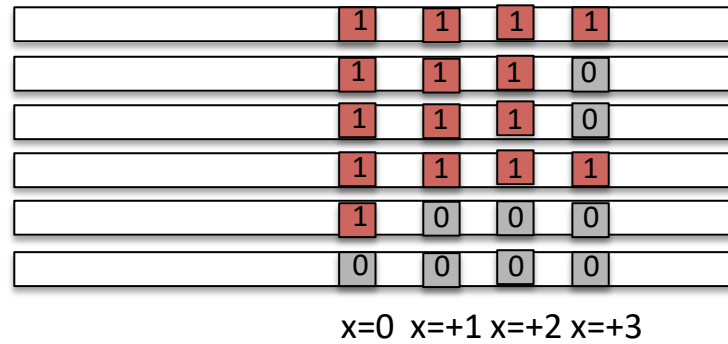


$$EHH_c(x_i) = \sum_{h \in H_c(x_i)} \frac{\binom{n_h}{2}}{\binom{n_c}{2}}$$

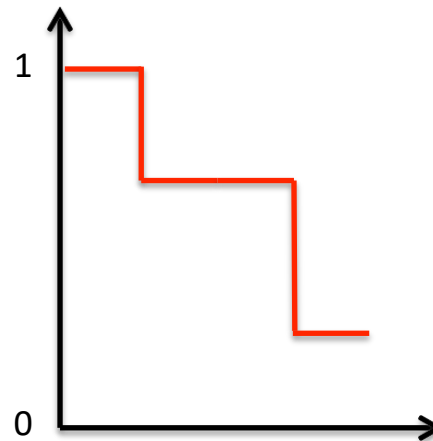


How many unique haplotypes carrying the core SNP?
What is their frequency?

Extended Haplotype Homozygosity (EHH)

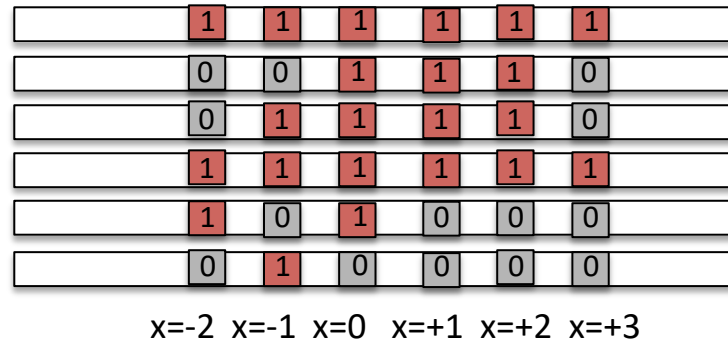


$$EHH_c(x_i) = \sum_{h \in H_c(x_i)} \frac{\binom{n_h}{2}}{\binom{n_c}{2}}$$

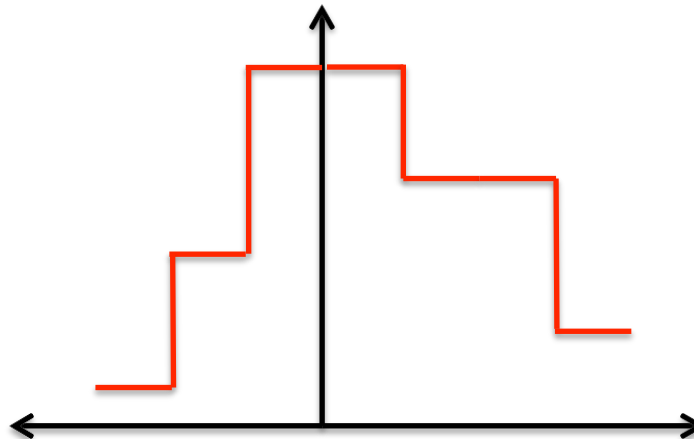


$$EHH_c(x_i = +3) = \frac{\binom{2}{2} + \binom{2}{2} + \binom{1}{2}}{\binom{5}{2}} = \frac{1+1+0}{10} = 0.20$$

Extended Haplotype Homozygosity (EHH)

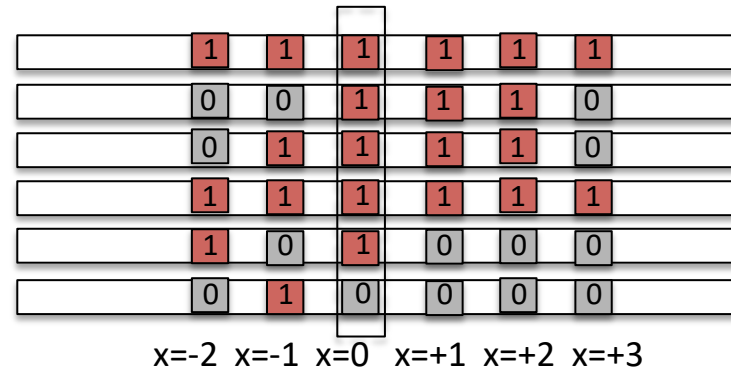


$$EHH_c(x_i) = \sum_{h \in H_c(x_i)} \frac{\binom{n_h}{2}}{\binom{n_c}{2}}$$

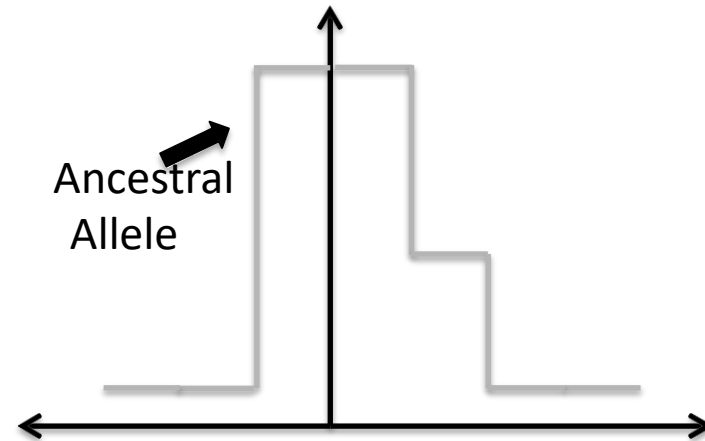
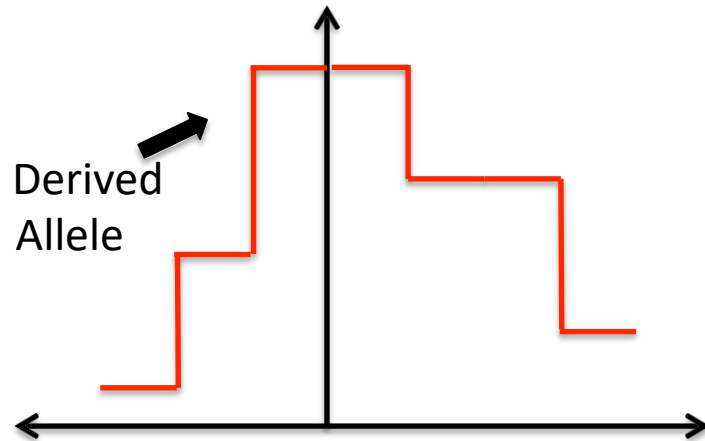


$$EHH_c(x_i = -1) = \frac{\binom{3}{2} + \binom{2}{2}}{\binom{5}{2}} = \frac{3+1}{10} = 0.4$$

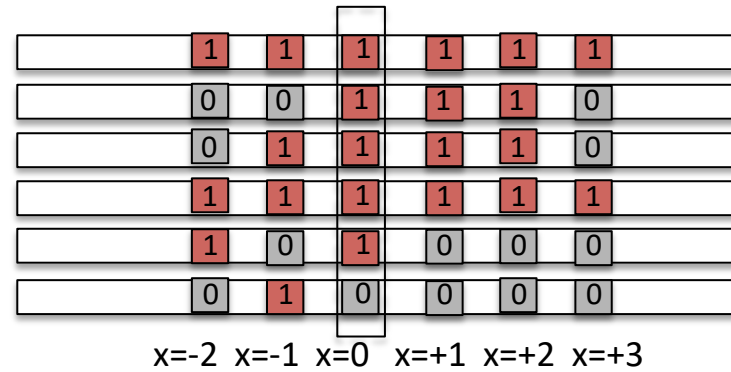
Extended Haplotype Homozygosity (EHH)



$$EHH_c(x_i) = \sum_{h \in H_c(x_i)} \frac{\binom{n_h}{2}}{\binom{n_c}{2}}$$

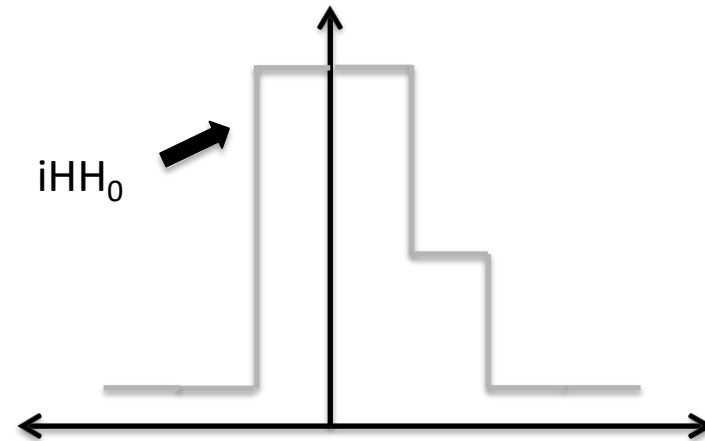
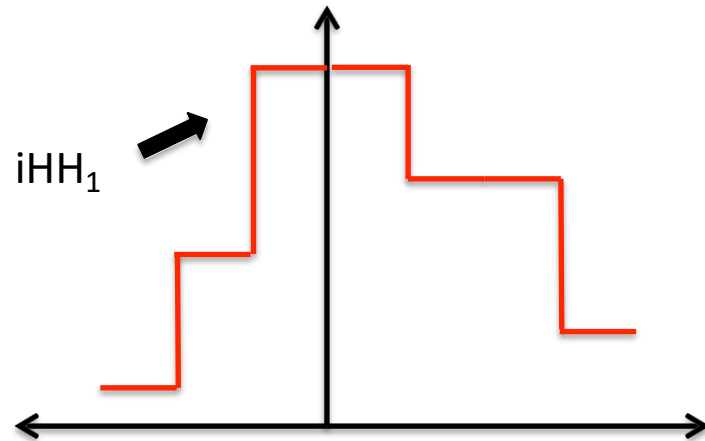


Integrated Haplotype Score (iHS)



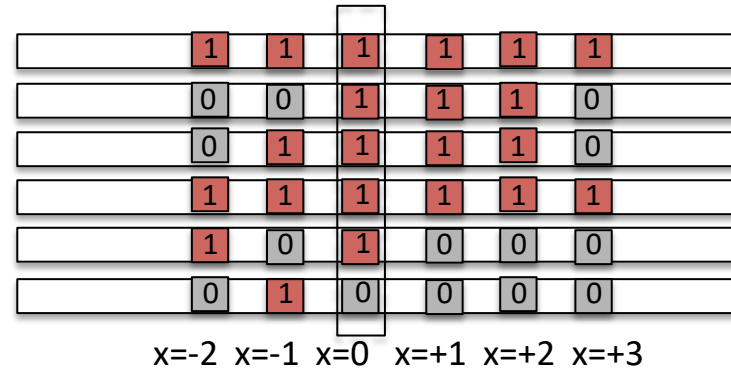
Integrated Haplotype Homozygosity (iHH)

$$iHS = \ln(iHH_1 / iHH_0)$$



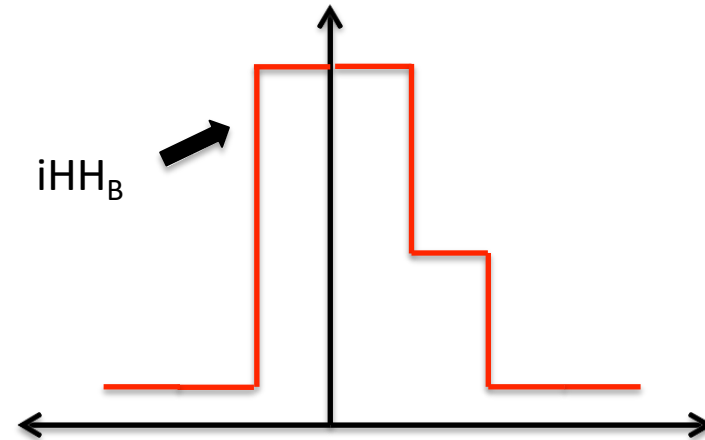
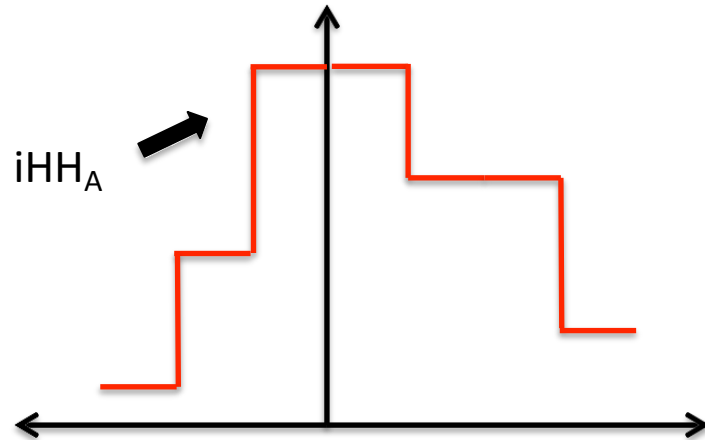
< -2 = derived allele
> 2 = ancestral allele

Cross Population Extended Homozygosity Haplotype (xpEHH)



Integrated Haplotype Homozygosity
(iHH) for A and B

$$xpEHH = \ln(iHH_A / iHH_B)$$



Recent Advances to Detect Selection

1. Composite scores (Grossman et al. 2013)

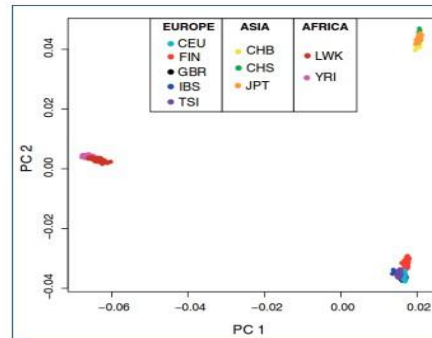
$$BF_t = \frac{P(v_t \in \text{bin}_{t,k} | \text{selected})}{P(v_t \in \text{bin}_{t,k} | \text{unselected})}$$

and defined the composite score as the product of the Bayes factor of each test:

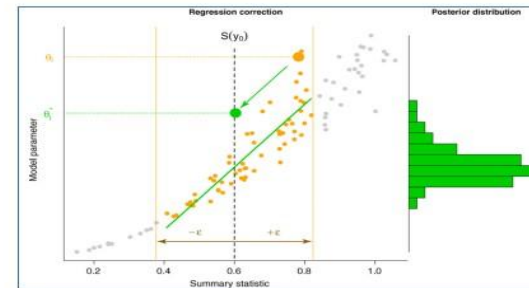
$$CMS_{GW} = \prod_{t \in \text{tests}} BF_t$$

3. Unsupervised machine learning

(PCA, Duforet-Frebourg et al. 2016)

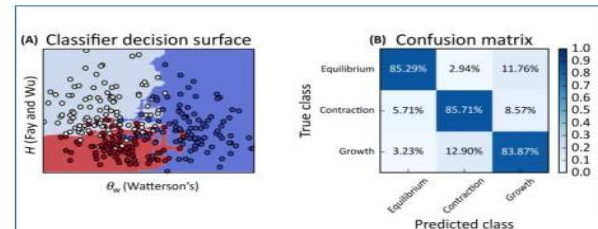


2. Simulations-based (rejection, ABC)



4. Supervised machine learning

(SVM, Schrider & Kern 2018)



Thank you!

